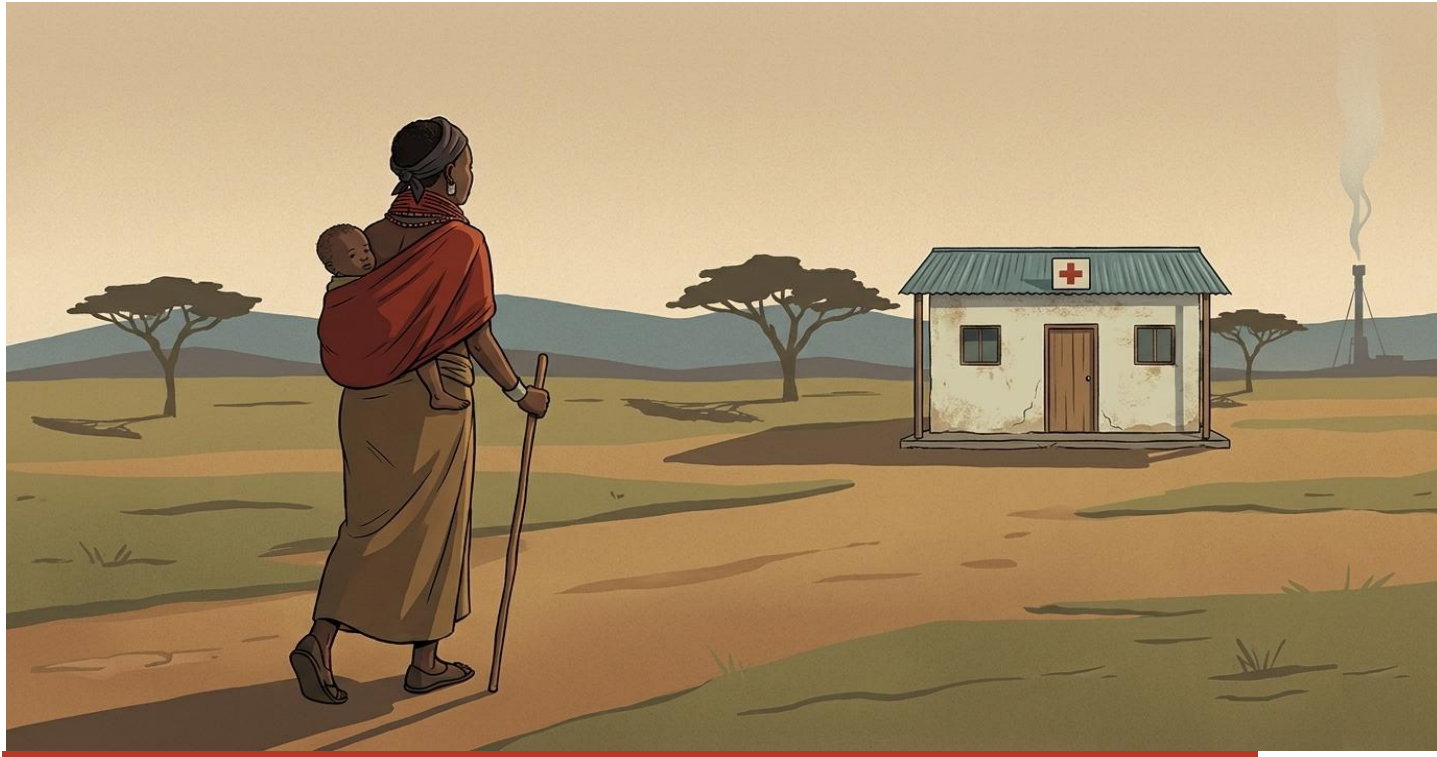




A PORTFOLIO PIECE BY ERIC KIZAMBO KACHILA

Baringo-Silali Community Health Access Programme

Impact Evaluation Design

A quasi-experimental impact evaluation design for a CSR-funded community health programme serving remote pastoralist communities in Baringo County, Kenya.

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About this document

This portfolio piece comprises two related artefacts developed around a single intervention — the Baringo-Silali Community Health Access Programme (BSCHAP) — to demonstrate complementary evaluation competencies. Part I is an Impact Evaluation Design Document, written in the evaluator's voice, that develops a complete quasi-experimental and mixed-methods evaluation design from intervention selection through to dissemination and use. Part II is a standalone Terms of Reference, written in the institutional voice of a commissioning body, that demonstrates the capacity to produce the document that would initiate such an evaluation. The two documents reference each other throughout and are designed to be read as a single coherent body of work.

The implementing agency is anonymized throughout; the geographic and programmatic context of Baringo County, Kenya, is retained for analytical realism. All data and figures presented are illustrative.

Section 1: Programme and Evaluation Scope

1.1. Programme background and rationale

Baringo-Silali is among Kenya's most geographically isolated and structurally underserved regions. Located within Baringo County in the northern Rift Valley, the area is characterized by sparse population, arid and semi-arid terrain, weak road infrastructure, and an almost complete absence of state delivered social services. For the pastoralist and agro-pastoralist communities that inhabit the area, the nearest public health facility is several hours away by foot or motorcycle, and during rainy seasons becomes effectively unreachable. Routine primary care, antenatal services, and emergency response are, for practical purposes, unavailable.

A geothermal energy developer operating a project site in the area established a small clinic on the project camp to provide occupational health services to its workforce. Recognizing the depth of the access gap in surrounding communities and the absence of any plausible alternative provider, the developer extended the clinic's services to community members under its Corporate Social Responsibility framework. The Baringo-Silali Community Health Access Programme (BSCHAP) provides outpatient consultations, maternal and child health services, basic diagnostics, immunizations, and emergency first aid free of charge to community members residing within approximately thirty kilometers of the project camp.

BSCHAP has been operational for several years and represents the only consistent source of formal primary healthcare for the communities it serves. It is, in essence, a non-state actor filling a state-provision gap. Despite its operational maturity, no structured evaluation of its effect on community health outcomes has been undertaken. This evaluation design responds to that gap. It treats BSCHAP as a discrete, evaluable component of the implementing agency's broader community engagement portfolio and proposes a quasi-experimental design appropriate to the programme's retrospective and observational character. The selection of BSCHAP as the evaluable unit rather than the wider energy development project of which it is a part reflects the standard impact evaluation criterion that interventions with identifiable individual beneficiaries, plausible counterfactuals, and attributable treatment effects are more credibly evaluated than infrastructure programmes whose impacts operate at institutional or grid level.

1.2. Intervention description and evaluation scope

BSCHAP delivers a defined set of primary healthcare services to community members residing within the catchment of the project camp clinic. Service components include outpatient consultations for acute and chronic conditions, antenatal and postnatal care, child immunizations, basic diagnostic testing, dispensing of essential medicines, and referral to higher-level facilities for cases beyond the clinic's scope of practice.

Services are provided free of charge to community beneficiaries, distinguishing the programme from the cost-recovery models that characterize much of Kenya's faith-based and private primary care sector.

The intervention is implemented by the CSR function of the geothermal energy developer (hereafter referred to as *the implementing agency*) in Baringo County, Kenya. The intervention's reference period for this evaluation is 2020 to present. Financing is drawn from the implementing agency's annual CSR allocation and covers clinical personnel, medical supplies, diagnostic consumables, and facility operating costs. Specific budget figures are not reproduced in this document; the order of magnitude is consistent with a single-facility primary care operation in a low-income, low-cost rural Kenyan setting.

The evaluation scope is deliberately bounded. The component to be evaluated is the Community Outpatient Health Access Service, i.e. the provision of free primary healthcare to members of communities within approximately thirty kilometers of the project camp. Occupational health services delivered to the implementing agency's workforce are excluded from the evaluation, as are broader community engagement activities (water provision, education support, and livelihoods programming) that may run in parallel under the wider CSR portfolio. This bounding is methodologically deliberate: by isolating a single, well-defined service-delivery intervention, the evaluation preserves the conditions under which causal claims can be defended and avoids the attribution problems that arise when multiple co-located interventions are treated as a single bundle.

1.3. Programme objectives and intervention logic

The programme is organized around four objectives. The evaluation focuses primarily on the first, which is the most directly attributable to the service-delivery intervention and the most amenable to counterfactual reasoning.

- Improve access to primary healthcare for remote communities surrounding Baringo-Silali that have no alternative health facility within reasonable reach.
- Reduce morbidity and preventable illness in the catchment population through earlier diagnosis and treatment of common conditions.
- Improve maternal and child health outcomes through antenatal care, immunisation, safe delivery support, and postnatal follow-up.
- Reduce financial hardship associated with seeking healthcare by removing direct service costs and reducing travel costs to distant facilities.

The selected evaluable objective for this design is the first improved access to primary healthcare and the downstream effects on treatment-seeking behaviour, health outcomes, and household financial protection. This objective satisfies the criteria that justify an impact evaluation: the beneficiary population

is identifiable at the individual and household level; the treatment is discrete and attributable; comparable communities without clinic access exist and can serve as a counterfactual; and the relevant outcomes are measurable through standard household survey instruments.

The intervention's logic, in its most compact form, can be expressed as a sequence from inputs to ultimate outcomes. The full theory of change and disaggregated results chain are developed in Section 2 of this design document; the summary here is intended only to orient the reader to the causal logic that the evaluation will test.

- **Inputs:** clinical personnel, medicines and consumables, diagnostic equipment, the clinic facility itself, CSR financing, and community outreach logistics.
- **Activities:** outpatient consultation, antenatal and postnatal care provision, immunization campaigns, dispensing, referral, and community health education.
- **Outputs:** consultations delivered, ANC and PNC visits conducted, immunizations administered, referrals made, and medicines dispensed.
- **Immediate outcomes:** increased formal treatment-seeking, earlier diagnosis, increased uptake of antenatal and immunization services, reduced out-of-pocket transport costs.
- **Ultimate outcomes (impact):** reduced under-5 mortality, increased skilled birth attendance, reduced incidence of preventable illness, and reduced incidence of catastrophic household health expenditure.

1.4. Target population and beneficiaries

The direct beneficiaries of BSCHAP are community members residing in approximately eight to twelve villages located within a thirty-kilometer radius of the project camp clinic. Clinic registers indicate that between five hundred and eight hundred unique community individuals access the clinic in any given year, with cumulative beneficiaries since community access began estimated to exceed two thousand individuals. These registers, while not designed as evaluation instruments, provide the basis for retrospective enumeration of the treated population and for establishing service utilization trajectories over time.

The catchment population is predominantly drawn from the Pokot and Tugen ethnic communities. Livelihoods are dominated by pastoralism and subsistence agriculture, with limited cash-economy activity. Household incomes are low by national standards and highly seasonal. Educational attainment is below the national average, and women and children comprise the majority of clinic users, a pattern consistent both with the gendered division of healthcare-seeking labour in pastoralist contexts and with the maternal and child health emphasis of the service offering.

For analytical purposes, the evaluation will disaggregate beneficiaries along four dimensions: sex, age band (with explicit attention to women of reproductive age and children under five), livelihood category (pastoralist versus agro-pastoralist), and distance from the clinic. The first three dimensions matter because the intervention is hypothesized to have heterogeneous effects across population sub-groups, for example, larger effects on maternal and child outcomes than on adult morbidity, given the service mix. The fourth dimension, distance matters because access cost is a key mechanism through which the clinic is expected to operate, and effects are expected to attenuate with distance. The evaluation's heterogeneity analysis is built around these four dimensions.

Beyond direct beneficiaries, BSCHAP generates indirect benefits to all households in the catchment communities through the financial protection afforded by free service provision, the productivity gains associated with reduced illness in working-age adults, and the broader normalization of formal health-seeking behavior. These indirect effects are difficult to attribute cleanly and will not be the focus of the primary causal analysis but will be discussed in the evaluation's interpretive narrative.

1.5. Evaluation question and outcome indicators

The evaluation is framed as a magnitude-estimation rather than hypothesis-testing exercise. In a setting where the baseline absence of primary healthcare is well documented and where the intervention provides a service that is, in standard public health terms, almost certain to produce some positive effect, the policy-relevant question is not whether BSCHAP improves health outcomes but by how much, for whom, and at what cost. Framing the question as binary, does the programme work, would yield a finding whose answer was largely known in advance and would not support the resource allocation decisions that the evaluation is intended to inform. Magnitude framing, by contrast, generates effect-size estimates that can be compared against the implementing agency's investment, against benchmark cost-effectiveness ranges for primary care interventions in similar settings, and against alternative uses of CSR resources.

Primary causal question

What is the magnitude of the effect of BSCHAP clinic access on (a) formal treatment-seeking behavior, (b) skilled birth attendance and antenatal care completion, and (c) household out-of-pocket health expenditure, among rural pastoralist communities within thirty kilometres of the Baringo-Silali project site — and how does this effect vary across women of reproductive age, children under five, and households at varying distances from the clinic?

Secondary policy question

Is the estimated impact per CSR shilling invested in BSCHAP comparable to benchmark cost-effectiveness ranges for primary healthcare access interventions in similar low-resource African settings, and does the magnitude of effect justify continued or expanded investment?

Outcome indicators

The evaluation tracks seven indicators across the results chain. Three are positioned at the immediate outcome level capturing behavioral and service-uptake changes that are expected to occur in the short-term following the establishment of clinic access. Four are positioned at the ultimate outcome (impact) level capturing the durable changes in population health and household economic security that constitute the programme's reason for existing. The distinction between immediate outcomes and impact-level indicators is methodologically important: behaviour change and service uptake are necessary but not sufficient evidence of impact, and an evaluation that conflates the two cannot speak credibly to whether the programme has produced the changes it was designed to produce.

Indicator	Level in results chain	Unit / measure	Data source
Treatment-seeking rate	Immediate outcome	% of households reporting formal care-seeking in last episode of illness	Household survey (treatment + comparison)
Antenatal care completion	Immediate outcome	% of pregnant women completing 4+ ANC visits	Clinic registers + household survey
Full childhood immunization	Immediate outcome	% of children under 5 fully immunized per national schedule	Clinic registers + household survey
Skilled birth attendance	Ultimate outcome / impact	% of live births attended by skilled health personnel	Household survey + facility records
Under-5 mortality rate	Ultimate outcome / impact	Deaths per 1,000 live births among children under 5	Household survey (retrospective birth histories)
Incidence of preventable illness	Ultimate outcome / impact	Episodes of preventable illness per household per year	Household survey (recall) + clinic records

Catastrophic health expenditure	Ultimate outcome / impact	% of households with health spending exceeding 10% of household income	Household expenditure survey
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All indicators will be measured at the household level through a structured survey administered in both treatment and comparison communities. Clinic registers will supplement household-reported data for service-utilisation indicators, providing a triangulation source that strengthens internal validity. The detailed sampling strategy, instrument design, and power calculations are presented in Section 4 of this design document.

1.6. Intended users and use of the evaluation

An impact evaluation generates value only to the extent that its findings are used. The design of this evaluation is therefore deliberately oriented toward four user groups, each of which faces a distinct decision that the evaluation can inform. The framing of the primary and secondary evaluation questions, the choice of outcome indicators, and the dissemination strategy are all calibrated to the needs of these users.

Intended user	Decision the evaluation will inform
CSR Department (implementing agency)	Whether to sustain, expand, or restructure community health investment; what indicators to monitor going forward.
Senior Management (implementing agency)	Strategic positioning of community health within the wider CSR portfolio; communication of impact to shareholders and regulators.
County Government of Baringo (Department of Health)	Whether to formalize referral linkages with BSCHAP, integrate the clinic into county health planning, or replicate the model in other underserved wards.
Development partners and INGOs operating in arid and semi-arid lands	Comparative evidence on CSR-funded primary care models as a service-delivery modality in contexts where the state is absent.

The CSR Department of the implementing agency is the primary user. This department commissions and funds the programme and is accountable, internally and to senior management, for the value generated by its CSR investments. Magnitude estimates and cost-effectiveness comparisons will directly inform its decisions about programme continuation, expansion, and benchmarking. Senior management is a secondary user, with interest in the strategic and reputational dimensions of the evaluation findings.

The county government's Department of Health is positioned as an external user with operational relevance. Where BSCHAP demonstrates significant effects, the case for formalizing referral pathways and integrating the clinic into county health planning strengthens. The evaluation findings are therefore designed to be intelligible to a county public health audience and to support a conversation about partnership and replication.

Development partners and international NGOs operating in Kenya's arid and semi-arid lands are positioned as a fourth user group. CSR-funded primary care models occupy an unusual position in the development financing landscape — they are neither state provision, nor donor-funded humanitarian response, nor private sector market activity. Comparative evidence on whether such models can produce measurable health effects at meaningful magnitudes contributes to a wider conversation about service-delivery modalities in contexts where the state is structurally absent. The evaluation is designed to be publishable in a form that supports that conversation.

Finally, the evaluation itself is intended to model good practice in the evaluation of CSR-funded interventions — a category of programming that is widely undertaken but rarely evaluated to the standard applied to publicly or donor-funded equivalents. The design choices documented here are made with that wider audience in mind.

Section 2: Theory of Change and Results Framework

2.1. Purpose of the theory of change

A theory of change makes explicit the causal logic by which an intervention is expected to produce its intended effects. For an impact evaluation, the theory of change serves three distinct functions. First, it identifies the points along the causal chain at which the intervention is expected to produce measurable change, and therefore the indicators that the evaluation must capture. Second, it surfaces the assumptions on which the causal claim rests — the conditions that must hold, but lie outside the programme's direct control, for the intended chain of effects to be realised. An evaluation that takes a programme's theory at face value, without examining its assumptions, cannot interpret a null or weak result; the finding could mean the programme does not work, or it could mean that an underlying assumption did not hold. Third, the theory of change distinguishes the intervention's mechanism — how it works — from the external conditions under which the mechanism produces impact. This distinction matters for transferability: a programme that works in one context may fail in another not because its mechanism is invalid, but because a development assumption that held in the first context does not hold in the second.

The theory of change presented here is developed retrospectively. BSCHAP was not designed as a research intervention with a pre-specified causal model; it was established as an operational response to a documented access gap, and the causal logic has been reconstructed for evaluation purposes. This is a common feature of impact evaluations of established programmes and is methodologically defensible, provided the reconstruction is transparent about the evidence base on which each link in the chain is justified.

2.2. The BSCHAP theory of change

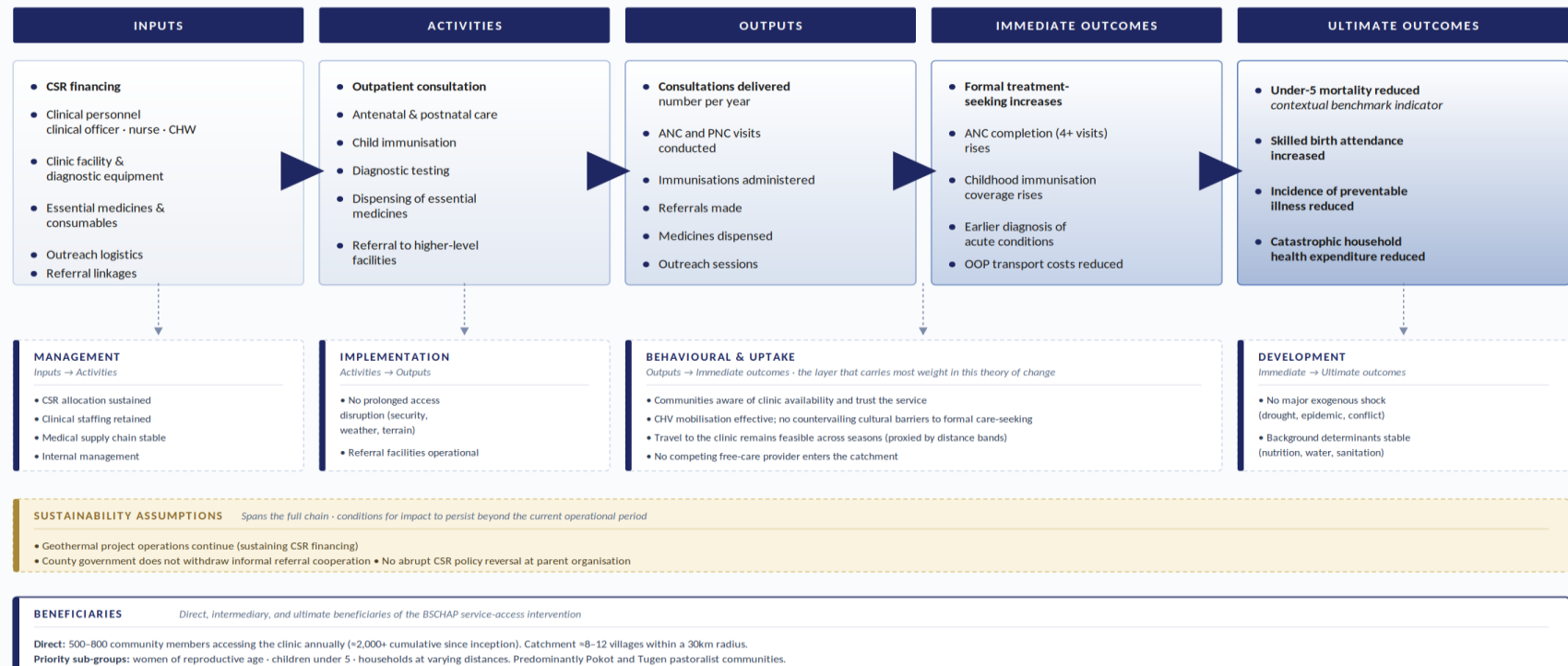
BSCHAP's theory of change is a service-access model. The programme rests on the proposition that the absence of formal primary healthcare in remote pastoralist communities is a binding constraint on a wide range of health outcomes, and that the introduction of a free, accessible, professionally staffed clinic relaxes this constraint with detectable downstream effects on treatment-seeking behaviour, service uptake, morbidity, mortality, and household financial security.

Figure 1 on the following page presents the full theory of change as a visual logic model, showing the results chain from inputs to ultimate outcomes alongside the four assumption layers and the sustainability conditions that span the chain. The narrative that follows the figure develops each component.

FIGURE 1 - THEORY OF CHANGE

Baringo-Silali Community Health Access Programme

Results chain from inputs to ultimate outcomes, with four assumption layers and sustainability conditions



Note: Assumption layers follow the standard logic-model convention. Items that are mechanistic in nature (i.e. statements of how the intervention works) are addressed in the theory-of-change narrative in Section 2.2, not as assumptions. The diagram retains only genuinely external enabling conditions at each layer.

BSCHAP IMPACT EVALUATION DESIGN · SECTION 2 · EVALUATION DESIGNER: ERIC KIZAMBO KACHILA

Figure 1. BSCHAP Theory of Change and Results Chain

Source: Author's reconstruction based on programme documentation and field observation.

The inputs to BSCHAP are the resources that the implementing agency commits to the programme through its CSR allocation. These include the operating budget, the clinical personnel based at the project camp, the clinic facility and its diagnostic equipment, the medicines and consumables consumed in service delivery, the outreach logistics that connect the clinic to surrounding villages, and the referral linkages with county-level facilities that handle cases beyond the clinic's clinical scope. The combination of these inputs is, in operational terms, modest — BSCHAP is a single-facility primary care operation, not a complex multi-component intervention. The simplicity of the input structure is a methodological asset: it makes the treatment well-defined and the attribution problem more tractable.

Activities are the operations through which the inputs are converted into services. Outpatient consultation is the primary activity by volume; maternal and child health services (antenatal care, postnatal care, immunisation) constitute the second activity stream and are the basis for several of the evaluation's outcome indicators. Diagnostic testing and dispensing are supporting activities; referral to higher-level facilities and community outreach extend the programme's reach beyond the clinic walls.

Outputs are the direct products of these activities, expressed as counts of service-delivery transactions: consultations delivered, ANC and PNC visits conducted, immunisations administered, referrals made, medicines dispensed, and outreach sessions conducted. An important conceptual point is captured in the diagram: output counts already imply uptake, in the sense that a consultation cannot be 'delivered' to a person who is not present. This means that the question of whether communities access the clinic is not, strictly, an assumption operating between activities and outputs — it is a constituent feature of the outputs themselves. The relevant uptake-related assumptions operate one step further along the chain, between outputs and immediate outcomes, and concern whether the population-level rates of behavioural change implied by service delivery are realised. This is the behavioural and uptake assumption layer in Figure 1.

Immediate outcomes are the short-term changes in population-level behaviour and uptake that follow from sustained service delivery: a rise in formal treatment-seeking when household members fall ill, an increase in the proportion of pregnancies that complete four or more antenatal visits, a rise in full childhood immunisation coverage, earlier diagnosis of acute conditions because care is geographically reachable, and a reduction in the out-of-pocket transport costs that previously deterred or financially burdened care-seeking. These outcomes are the principal targets of the impact evaluation: they are the points at which the programme's effect is most likely to be detectable, most clearly attributable, and most adequately powered given the catchment population.

Ultimate outcomes are the durable population-level health and financial-protection effects that constitute the programme's reason for existing. Reductions in under-5 mortality and the incidence of preventable illness, increases in skilled birth attendance, and reductions in catastrophic household health

expenditure all sit at this level. These are the outcomes that ultimately justify the investment, but they are also the most distant from the intervention in time and causal sequence and therefore the most subject to confounding influences. The evaluation's analytical strategy for these indicators is more cautious than for the immediate outcomes — a point developed in Section 2.4 below and treated in detail in Section 4.

2.3. Assumptions: what must hold for the chain to work

Each transition along the results chain depends on conditions that lie outside the direct control of the programme. The four assumption layers shown in Figure 1 — management, implementation, behavioural and uptake, and development — partition these conditions according to the link of the chain at which they operate. A fifth layer, sustainability assumptions, spans the full chain and concerns whether the realised effects persist beyond the current operational period.

The distinction between assumptions and mechanisms is important and frequently elided. A statement such as "earlier diagnosis reduces severity of disease" is not an assumption; it is the causal mechanism through which the intervention operates. Assumptions are conditions external to the mechanism — they specify the environment in which the mechanism can operate. A development assumption that 'no major drought displaces the catchment population during the evaluation period' is genuinely external; a statement that 'antenatal care reduces maternal risk' is internal to the mechanism. This distinction has been applied in constructing the assumption layers presented below, with the consequence that the development assumption layer is leaner and more narrowly scoped than is sometimes seen in programmatic logic models.

Layer	Position in the chain	Examples for BSCHAP	Why this layer matters
Management assumptions	Inputs → Activities	CSR allocation sustained at adequate level; clinical staffing retained at the site; medical supply chain stable and uninterrupted.	Internal organisational conditions enabling resources to be converted into operational activity.
Implementation assumptions	Activities → Outputs	No prolonged disruption to clinic access (security, weather, terrain); referral facilities at county level operational and accepting BSCHAP referrals.	Operational conditions enabling activities to produce the intended service-delivery counts.

Behavioural and uptake assumptions	Outputs → Immediate outcomes	Communities aware of clinic availability and trusting of the service; CHV mobilisation effective; no countervailing cultural barriers to formal care-seeking; travel feasible across seasons; no competing free-care provider entering the catchment.	External conditions enabling service delivery to translate into uptake and behavioural change at the population level. This layer is explicit because uptake is not automatic.
Development assumptions	Immediate outcomes → Ultimate outcomes	No major exogenous health shock (drought, epidemic, conflict) affecting the catchment; background determinants of health (nutrition, water, sanitation) stable or improving in parallel; no countervailing health-degrading trend.	Genuinely external enabling conditions for short-term behavioural change to translate into durable population-level impact. Distinct from the causal mechanism itself.
Sustainability assumptions	Spans the full chain	Geothermal project operations continue (sustaining CSR financing); county government does not withdraw informal referral cooperation; no abrupt CSR policy reversal at parent organisation.	Conditions enabling programme impact to persist beyond the current operational period, relevant for interpreting evaluation findings as transferable.

Several implications follow from the assumption structure presented above, two of which are worth flagging at this stage. First, the behavioural and uptake assumption layer carries substantial weight in this theory of change. BSCHAP is a supply-side intervention — it provides a service — but its effects depend critically on demand-side uptake. The evaluation must therefore be sensitive to the possibility that null or weak effects on immediate outcomes reflect uptake failure rather than service-delivery failure. Second, the sustainability assumption that geothermal project operations continue carries unusual weight here: BSCHAP is funded from the CSR allocation of an active project site, and a decision by the implementing agency to wind down or relocate the project would directly remove the programme's financial basis. This is an unusual structural feature of CSR-funded interventions and is explicitly recognised in the evaluation's interpretive framing.

2.4. Results chain and outcome indicators

Table 1 below restates the full results chain in tabular form, providing a reference structure for the indicator placement that follows. Each stage of the chain is paired with a brief description of what it captures and the specific examples that apply to BSCHAP.

Table 1. Results chain for BSCHAP

Stage	What it captures	Examples for BSCHAP
Inputs	Resources committed to the intervention	CSR financing; clinical personnel (clinical officer, nurse, CHW); clinic facility and diagnostic equipment; essential medicines and consumables; outreach logistics; referral linkages.
Activities	What the programme does with its inputs	Outpatient consultation; antenatal and postnatal care; child immunization; diagnostic testing; dispensing of essential medicines; referral to higher-level facilities; community outreach.
Outputs	Direct products and services produced by activities, expressed as counts of service-delivery transactions	Consultations delivered; ANC and PNC visits conducted; immunizations administered; referrals made; medicines dispensed; outreach sessions conducted.
Immediate outcomes	Short-term behavioral and uptake changes among the catchment population, conditional on outputs occurring and being taken up	Increase in formal treatment-seeking; rise in completion of four or more antenatal visits; rise in full childhood immunization; earlier diagnosis of acute conditions; reduction in household out-of-pocket transport costs.
Ultimate outcomes (impact)	Durable, population-level health and financial-protection outcomes that the intervention is intended to produce	Reduction in under-5 mortality; increase in skilled birth attendance; reduction in incidence of preventable illness; reduction in catastrophic household health expenditure.

The seven indicators introduced in Section 1 are placed at the immediate-outcome and ultimate-outcome (impact) levels of the chain. Three indicators sit at the immediate-outcome level: treatment-seeking behaviour, antenatal care completion, and full childhood immunisation. These indicators capture the short-term behavioural and uptake changes that the programme is most directly positioned to produce and that, conditional on a credible counterfactual, can be most cleanly attributed to it. Four indicators sit at the ultimate-outcome level: under-5 mortality, skilled birth attendance, incidence of preventable

illness, and catastrophic household health expenditure. These indicators capture the durable population-level outcomes that justify the programme's existence but that operate at a greater causal distance from the intervention and depend on more assumptions holding.

2.5. Indicator feasibility and the analytical strategy

An indicator that is conceptually appropriate is not necessarily analytically usable. The choice of which indicators to treat as primary quantitative measures, which to present descriptively, and which to use as contextual benchmarks must be informed by the catchment population's size and the underlying rate at which the relevant events occur.

BSCHAP serves an estimated 500 to 800 unique community members per year, with a cumulative beneficiary count of approximately 2,000 individuals since community access began. This population is, in the language of evaluation design, small. For indicators that measure relatively common behaviours or events — formal treatment-seeking when a household member falls ill, completion of four or more antenatal visits during pregnancy, household experience of catastrophic health expenditure — the catchment is sufficient to support estimation of treatment effects at standard confidence levels, particularly when supplemented by an appropriate comparison group and pooled across multi-year recall.

For indicators that measure rare events, the analytical situation is more challenging. Under-5 mortality is the clearest case. National and district-level under-5 mortality rates in similar Kenyan settings are of the order of forty to sixty deaths per thousand live births. In a catchment of several hundred individuals, the annual number of under-5 deaths is small in expectation, and the statistical noise around any estimated treatment effect is correspondingly large. The implication is not that under-5 mortality is irrelevant to the evaluation — it is the most consequential health outcome at stake and cannot be ignored — but that it cannot be the primary indicator on which causal claims are made. The evaluation's analytical strategy therefore treats under-5 mortality as a contextual indicator: it is reported descriptively, benchmarked against district and national rates, and used to characterise the population's health status and the plausibility of the broader effect pattern, but it is not used as the test of programme impact.

Table 2 makes the analytical treatment of each indicator explicit. This treatment is developed in full in Section 4 of this design document, where the sampling strategy, data collection plan, and power calculations are presented.

Table 2. Indicator-level analytical treatment

Indicator	Level	Analytical treatment in this evaluation
Treatment-seeking rate	Immediate outcome	Primary quantitative indicator. Sufficient household-level variance for difference-in-differences estimation.
Antenatal care completion (4+ visits)	Immediate outcome	Primary quantitative indicator. Pooled across multi-year recall to increase analytical sample.
Full childhood immunisation	Immediate outcome	Primary quantitative indicator. Coverage rates support inference at catchment level.
Skilled birth attendance	Ultimate outcome / impact	Quantitative indicator. Analysed using multi-year retrospective birth histories pooled across catchment villages.
Catastrophic health expenditure	Ultimate outcome / impact	Quantitative indicator. Household-level measure with adequate variance for inference.
Incidence of preventable illness	Ultimate outcome / impact	Quantitative indicator at household-episode level; presented with supporting clinical-register triangulation.
Under-5 mortality rate	Ultimate outcome / impact	Descriptive and contextual indicator. Small catchment population limits power for direct effect estimation; presented as a contextual benchmark against district and national rates.

One further point on indicator design merits brief mention. The seven indicators presented above are constructed to be measurable through a structured household survey, supplemented by triangulation against clinic registers for service-utilisation indicators. This choice is deliberate. Survey-measurable indicators avoid the cost and operational complexity of clinical measurement, biomarker collection, or facility-based observation systems that many health-sector evaluations require. The trade-off is between the higher measurement precision available through clinical instruments and the larger achievable sample size at given budget through household survey instruments. In the BSCHAP context, where the binding constraint is population size rather than measurement precision, the survey-based approach is preferred. The specific instruments, including the items that operationalise each indicator, are developed in the data collection plan in Section 4.

2.6. Linking the theory of change to the impact evaluation questions

Section 1 set out the evaluation's primary causal question — the magnitude of effect of BSCHAP clinic access on treatment-seeking behaviour, skilled birth attendance and antenatal care completion, and household out-of-pocket health expenditure, with heterogeneity analysis across women of reproductive age, children under five, and households at varying distances from the clinic. The theory of change developed here makes the structure of that question explicit. The three outcome domains in the primary question correspond to three of the immediate-outcome indicators (treatment-seeking, ANC completion, transport cost reduction) and one ultimate-outcome indicator (skilled birth attendance). The selection of these four outcomes reflects the analytical strategy described in Section 2.5: each is adequately powered for inference at the catchment scale, each is positioned at a point in the causal chain where attribution to the intervention is defensible, and each carries direct policy relevance for the intended users of the evaluation.

The full theory of change therefore provides the conceptual scaffolding for the evaluation. It identifies the indicators that the evaluation must measure, the assumptions whose validity must be examined when interpreting findings, the points along the causal chain at which the programme's effects are most likely to be detectable, and the boundaries of the causal claims that the evaluation can credibly support.

Section 3: Experimental Identification Strategies and Their Limits

3.1. Why assess experimental designs at all

An impact evaluation requires a credible counterfactual: an estimate of what the outcomes of beneficiaries would have been in the absence of the intervention. Experimental designs construct the counterfactual through random assignment, generating treatment and comparison groups that are, on average and at sufficient scale, statistically equivalent across all observed and unobserved characteristics. The randomisation is what licenses the causal claim: any post-intervention difference in outcomes between the groups can be attributed to the intervention itself rather than to pre-existing differences between the populations being compared.

Experimental designs are the methodological gold standard in the impact evaluation literature, and an assessment of their applicability is the first step in any rigorous identification strategy. Even where they are ultimately not feasible — as in the case of BSCHAP — the analytical work of considering them explicitly serves three purposes. It clarifies precisely what constraints make them unworkable, which in turn sharpens the choice of alternative method. It documents the methodological reasoning for users of the evaluation who may legitimately ask why a more rigorous design was not adopted. And it surfaces the trade-offs that any non-experimental design will need to address through other means.

This section examines three experimental variants in turn. Each is presented as it would operate under semi-imaginary pre-implementation conditions — that is, treating BSCHAP as if its design were still open to revision before service delivery began. This framing follows the convention of the impact evaluation literature, which assesses experimental designs against an idealised design moment that, in the case of BSCHAP, has already passed. The retrospective character of the assessment is itself one of the constraints the section identifies.

3.2. Randomised assignment (lottery design)

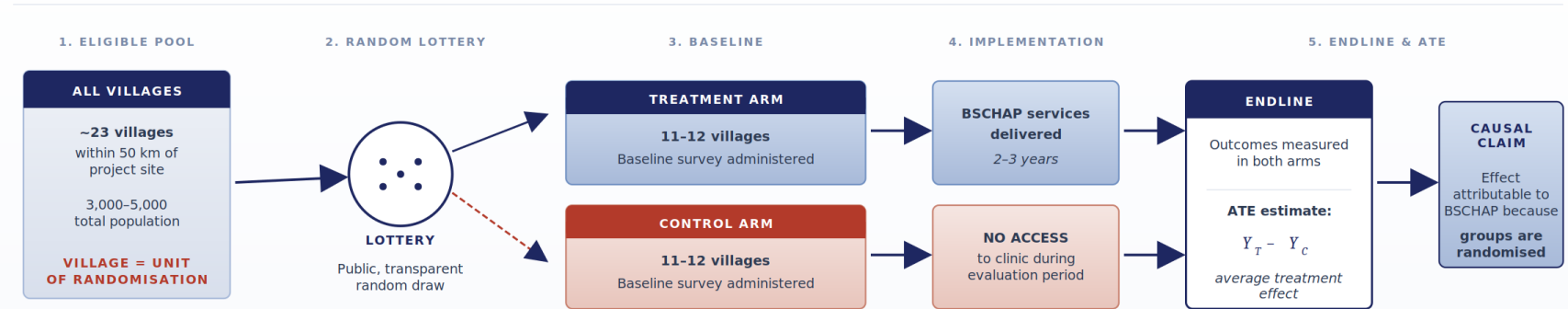
In a randomised assignment design, eligible units are randomly allocated either to receive the intervention or to be excluded from it during the evaluation period. Random allocation produces treatment and control arms that are, in expectation, equivalent on all baseline characteristics; the difference in mean outcomes between the two arms at endline identifies the average treatment effect (ATE). For BSCHAP, the natural unit of randomisation is the village, because the intervention operates at village level and households within the same village would be impossible to assign to different treatment statuses without contamination.

Figure 2 on the next page presents the design as it would operate if applied to BSCHAP. Approximately twenty-three villages within fifty kilometres of the project site would be enrolled. A public lottery would assign eleven to twelve villages to the treatment arm — receiving BSCHAP services immediately after the baseline survey — and the remaining eleven to twelve to the control arm, which would receive no clinic access during the two- to three-year evaluation period. Baseline measurement in both arms before treatment delivery, followed by endline measurement, would yield the difference-in-means estimate of programme impact.

FIGURE 2 · EXPERIMENTAL DESIGN — OPTION A

Randomised Assignment (Lottery Design)

All eligible villages are randomly assigned to either receive BSCHAP or to be withheld from access during the evaluation period.



WHY THIS DESIGN IS RULED OUT FOR BSCHAP

ETHICAL

Denying healthcare to control villages with no alternative facility violates non-maleficence. There is no genuine equipoise — primary care access is known to save lives. Withholding it for two to three years cannot be justified.

Informed consent from low-literacy pastoralist communities is procedurally and substantively problematic.

STATISTICAL — NUMEROSITY

With only ~23 villages randomised 11–12 per arm, the law of large numbers does not apply. Random assignment cannot reliably balance groups on observed or unobserved characteristics.

In practice the design would collapse into a difference-in-differences with baseline adjustment — which is the quasi-experimental approach developed in Section 4.

OPERATIONAL & REPUTATIONAL

The intervention is already operational; the design assumes a pre-implementation moment that no longer exists for BSCHAP.

If communities discovered that the implementing agency had deliberately withheld access from some villages for research, the social licence to operate would be at material risk.

VERDICT **RULED OUT.** Randomised assignment is ethically untenable, statistically underpowered, and operationally inapplicable to a pre-existing service-delivery intervention.

Figure 2. Randomised Assignment (Lottery Design) — design schematic and reasons for rejection in the BSCHAP context.

Source: Author's design, drawing on standard cluster-RCT conventions in impact evaluation.

The design is rejected on three grounds. The first is ethical and is, by itself, sufficient. In Baringo-Silali there is no alternative primary healthcare facility within reachable distance for the catchment population. Random assignment to the control arm therefore does not delay access or substitute one service for another; it constitutes permanent denial of a known-effective health intervention to communities that have no other option. The foundational principle of non-maleficence — do no harm — is breached when evaluation-related withholding of access leads to preventable illness, untreated emergency, or maternal death in control villages. There is, in clinical-trial terminology, no genuine equipoise: it is not in doubt that primary healthcare access saves lives, and randomising access to a known life-saving service cannot be justified by the value of the resulting evidence.

The second ground is statistical and concerns numerosity. Random assignment relies on the law of large numbers to deliver balanced groups: with enough units randomised to each arm, the average baseline characteristics of the two groups converge by construction. With approximately twenty-three villages randomised eleven to twelve per arm, the law of large numbers does not reliably apply. Random assignment at this scale can produce arms that differ systematically on observed or unobserved characteristics — wealth, distance from roads, livelihood composition, exposure to other services — purely by chance. The randomisation itself does not protect against such imbalances at small sample sizes. In practice, an experimental evaluation of BSCHAP would require baseline data on both arms and would collapse, analytically, into a difference-in-differences design with baseline adjustment. The randomisation would add little to a design that is already structurally non-experimental.

The third ground is operational and reputational. BSCHAP is already operational. The assessment of a randomised assignment design assumes a pre-implementation moment at which villages could still be allocated to arms; that moment has passed. Conducting such a design would require a substantial expansion of the programme — opening it to additional villages — combined with deliberate withholding of access from villages allocated to control. Beyond the ethical objection, the reputational risk to the implementing agency would be material if communities discovered that access had been withheld for research purposes, particularly if any adverse health outcome occurred in a control village during the evaluation period. The social licence to operate that the implementing agency depends on for the broader project is too consequential to put at this kind of risk for evidentiary gain that quasi-experimental methods can achieve at much lower social cost.

3.3. Randomized phase-in design

The randomized phase-in design responds to the ethical objection that the lottery design permanently denies treatment to control communities. In a phase-in, all units eventually receive the intervention, but the timing of access is randomised across cohorts. Comparison between cohorts that have received

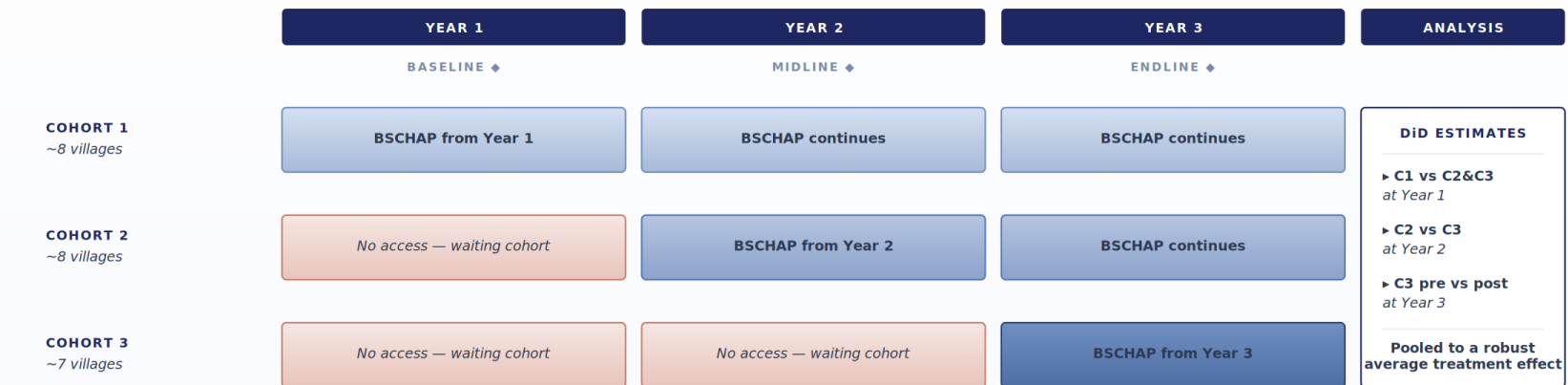
treatment and cohorts that have not yet received it generates a treatment effect estimate; the universal coverage at the end of the evaluation removes the equity objection that motivated the rejection of the lottery design.

Figure 3 presents the design as it would apply to BSCHAP. The twenty-three villages would be randomly assigned to three cohorts of roughly equal size. Cohort 1 would receive BSCHAP from Year 1, Cohort 2 from Year 2, and Cohort 3 from Year 3. Baseline, midline, and endline surveys would be conducted in all villages, generating three independent difference-in-differences comparisons — Cohort 1 against Cohorts 2 and 3 at Year 1, Cohort 2 against Cohort 3 at Year 2, and a pre–post comparison within Cohort 3 at Year 3 — which can be pooled to produce a robust average treatment effect.

FIGURE 3 · EXPERIMENTAL DESIGN — OPTION B

Randomised Phase-in Design

All villages eventually receive BSCHAP. Random assignment determines only the timing of access, in three sequential cohorts.



WHY THIS DESIGN IS RULED OUT FOR BSCHAP

ETHICAL — TIME IS NOT FUNGIBLE

Delayed access is not ethically neutral. A pregnant woman in a Year-3 cohort village cannot wait. Maternal deaths and untreated emergencies during the waiting period are irreversible.

No future access compensates for harm already incurred.

THE SAFEGUARDS PROBLEM

Adding ethical safeguards (emergency access, exemptions, CHV bridge) redefines what is being evaluated.

The contrast becomes **stronger vs weaker BSCHAP** — not BSCHAP vs no BSCHAP. The causal question that interests the implementing agency cannot be answered by this design.

STATISTICAL — NUMEROSITY

With ~23 villages split across three cohorts of 7–8, each phase comparison is severely underpowered.

Cohort imbalances on observable and unobservable characteristics cannot be ruled out by randomisation at this scale. Baseline data and matching would still be required.

VERDICT **RULED OUT.** Ethical safeguards required to deploy phase-in would redefine the treatment contrast away from the question the evaluation needs to answer.

BSCHAP IMPACT EVALUATION DESIGN · SECTION 3 · EVALUATION DESIGNER: ERIC KIZAMBO KACHILA

Figure 3. Randomised Phase-in Design — staggered cohort roll-out, difference-in-differences analysis across phase transitions.

Source: Author's design, following the phase-in conventions documented in Gertler et al. (2016).

The phase-in addresses the equity objection in principle but introduces three new problems in the BSCHAP context. The first is that the time fungibility assumption on which the design depends fails empirically. The design implicitly treats access in Year 2 as equivalent to access in Year 1 because all cohorts eventually receive the same programme; what is randomised is only when. This assumption holds when the relevant health outcomes are tolerant of delay — chronic conditions amenable to longer-term management, preventive services that produce benefits over years rather than weeks. It does not hold for the maternal, child health, and acute-care services that constitute the bulk of BSCHAP's outcome targets. A pregnant woman in a Cohort 3 village who experiences an obstetric emergency in Year 1 cannot wait two years for her cohort's turn. Whatever harm occurs during the waiting period is irreversible; no future access compensates for it. The phase-in does not solve the ethical problem; it postpones it by a year or two for some cohorts.

The second problem emerges if the design is rehabilitated by adding ethical safeguards — emergency-care access for waiting cohorts, exemptions for pregnant women in late stages of pregnancy, a community health volunteer bridge programme during the wait period, an independent ethics monitor with authority to halt the trial. Each of these safeguards is, individually, ethically necessary. Collectively, they redefine the treatment that is being evaluated. The contrast is no longer between full BSCHAP access and no BSCHAP access; it is between full BSCHAP and a partial bundle of BSCHAP-equivalent services delivered through the safeguard channels. The estimated treatment effect becomes the marginal effect of full versus partial service provision, which is a different and substantially less useful question for the implementing agency than the question the evaluation is intended to answer. A safeguarded phase-in could in principle be conducted, but it would not generate the evidence that the evaluation's intended users need.

The third problem is the same numerosity constraint that affects the lottery design. With twenty-three villages distributed across three cohorts of seven or eight, each phase transition produces an underpowered comparison. Imbalances on observed and unobserved characteristics cannot be ruled out by randomisation at this scale; baseline data and statistical adjustment for residual imbalance would be required in any case. As with the lottery design, the experimental rigour that motivates the choice of method does not, in practice, materialise.

3.4 Randomised promotion (encouragement design)

Randomised promotion, sometimes referred to as an encouragement design, differs from the two preceding options in a critical respect: it does not withhold the programme from anyone. The intervention is open to all units; what is randomised is the intensity of demand-side effort to encourage uptake. The design is used in settings where a programme is available but barriers to take-up exist — distance, cost,

awareness, cultural reluctance, mistrust — and where these barriers can be addressed through outreach, transport support, education, or peer encouragement. Random assignment of promotion intensity creates exogenous variation in uptake without restricting anyone's formal access to the programme.

This distinction matters for BSCHAP because the constraint that ruled out the lottery and phase-in designs — the ethical impermissibility of withholding healthcare — does not apply to a design that withholds only promotion. The conceptual clarification is significant: a household in a non-promoted village retains the same formal entitlement to BSCHAP services as a household in a promoted village. What differs is whether community health volunteers visit, whether transport subsidies are offered, whether trust-building sessions with elders are organised. The clinic doors are open to both arms; the demand-side support that helps households cross the threshold is what varies.

Figure 4 sets out the design. Approximately twenty-three villages would be randomly assigned to either a promotion arm — receiving CHV home visits, transport subsidies, trust-building sessions, and proactive appointment scheduling — or a no-promotion arm in which the clinic remains available but no active outreach is delivered. The promotion arm is expected to achieve higher take-up; the no-promotion arm provides the comparison level of organic uptake. Because the randomisation operates on promotion rather than on access, no community is excluded from the programme, and the ethical concerns that ruled out the previous two designs do not apply.

FIGURE 4 · EXPERIMENTAL DESIGN — OPTION C

Randomised Promotion (Encouragement) Design

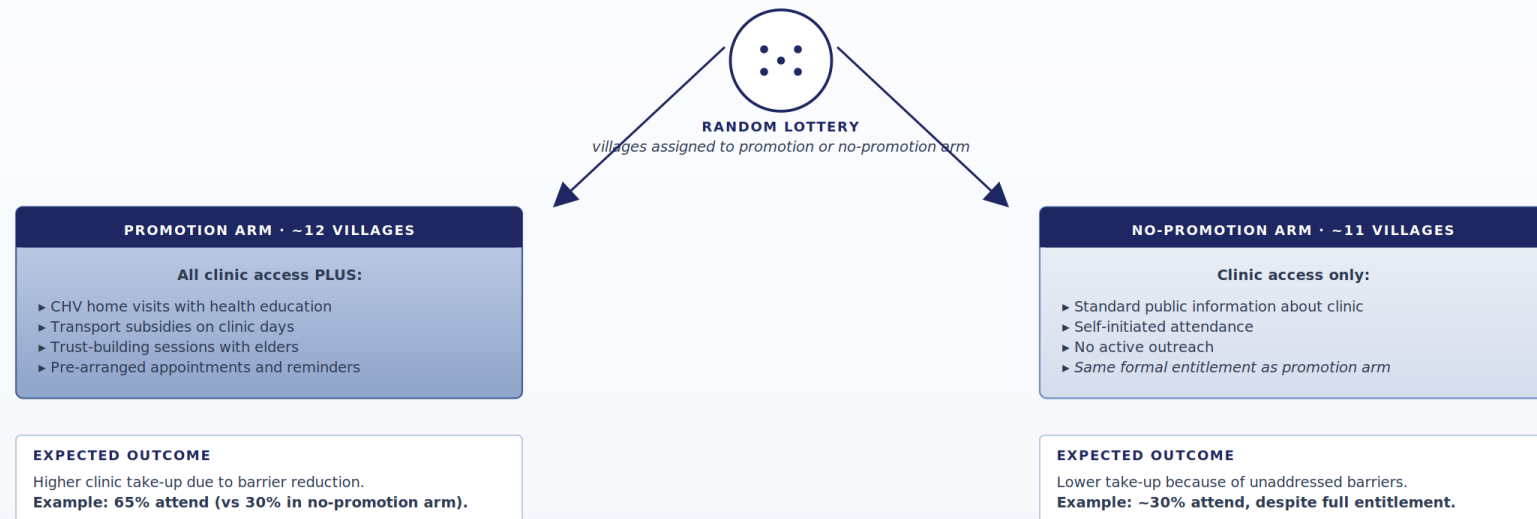
All villages have equal access to BSCHAP. Random assignment determines only which villages receive additional encouragement to overcome uptake barriers.

CONCEPTUAL CLARIFICATION

Randomised promotion does not withhold the programme from anyone. The clinic is open to all villages. What varies is the intensity of demand-side effort to overcome known barriers to uptake (e.g. mistrust, distance, awareness).

CLINIC ACCESS — UNIVERSAL

BSCHAP clinic open to all ~23 villages, free of charge, throughout the evaluation period.
Identical formal entitlement; differential take-up because of barriers.



LATE — LOCAL AVERAGE TREATMENT EFFECT · WALD ESTIMATOR

mean outcome (promoted) – mean outcome (not promoted)

$$LATE = \frac{\text{take-up rate (promoted)} - \text{take-up rate (not promoted)}}{\text{take-up rate (promoted)} - \text{take-up rate (not promoted)}}$$

Estimates the effect for compliers only —
those whose take-up was caused by the promotion.

BSCHAP IMPACT EVALUATION DESIGN · SECTION 3 · EVALUATION DESIGNER: ERIC KIZAMBO KACHILA

Figure 4. Randomised Promotion (Encouragement) Design — universal access with randomised demand-side support, identified through the Wald estimator.

Source: Author's design, following the encouragement-design framework in Duflo et al. (2008).

The analytical method associated with randomised promotion is instrumental-variables estimation. The promotion assignment serves as an instrument for actual programme participation: it is randomly assigned (and therefore independent of household characteristics), it affects participation (relevance), and it affects health outcomes only through participation (the exclusion restriction). Under these conditions, the ratio of the difference in mean outcomes between the two arms to the difference in take-up rates between the two arms — the Wald estimator — identifies the local average treatment effect (LATE): the causal effect of programme participation for the subset of households whose participation was actually caused by the promotion. The Wald form expresses the LATE in its most transparent way; two-stage least squares is the algebraic generalisation that becomes necessary once covariates or multiple instruments are introduced, but in the single-instrument case with no covariates the two reduce to the same number.

Two observations about LATE are important for interpreting what this design can and cannot tell the implementing agency. First, the LATE applies only to compliers — those whose take-up was caused by the promotion. It does not estimate the effect for always-takers (households that would attend regardless of promotion) or never-takers (households that would not attend even with promotion). Where the policy interest is in the population-average effect, the LATE may not be the parameter of interest. Second, the credibility of the design rests on the validity of the exclusion restriction: the requirement that promotion affects outcomes only through its effect on participation. If promotion itself produces health benefits — for example, through the health education content of CHV visits, independent of any clinic attendance it generates — the exclusion restriction is violated and the IV estimate is biased.

The randomised promotion design is therefore retained in this evaluation's identification strategy, but as a supplementary rather than primary approach. The intervention setting plausibly contains the kind of demand-side barriers that the design is built to address — mistrust of formal health providers, awareness gaps, transport costs, opportunity-cost of clinic visits — and the design provides a credible second source of identification that does not depend on the assumptions that the difference-in-differences approach in Section 4 relies on. Combined identification across methods that rely on different identifying assumptions is one of the most robust ways to defend a causal claim in a setting where no single method is unambiguously preferable. The role of randomised promotion in the overall identification strategy is developed in Section 4.

3.5 Summary and implications for the identification strategy

The assessment of the three experimental identification strategies leads to the conclusions summarised in Table 3 below. Randomised assignment and randomised phase-in are ruled out, on grounds that combine ethical considerations (denial of healthcare to communities with no alternative), statistical

considerations (insufficient numerosity to support credible randomisation), and operational considerations (the intervention is already in place). Randomised promotion is retained as a conditionally viable supplementary strategy that does not withhold the programme from anyone but generates exogenous variation in uptake through a different mechanism.

Table 3. Summary of experimental design assessment

Design option	Verdict	Why
Randomised assignment (lottery)	Ruled out	Withholding healthcare from communities with no alternative violates non-maleficence; with ~23 villages, randomisation cannot reliably balance groups; the intervention is already operational.
Randomised phase-in	Ruled out	Ethical safeguards required to make the design deployable would redefine the treatment contrast from 'BSCHAP vs no BSCHAP' to 'stronger vs weaker BSCHAP'; numerosity remains a binding constraint.
Randomised promotion (encouragement)	Conditionally viable	Does not withhold the programme. Could exogenously increase uptake by addressing known barriers (mistrust, awareness, transport). Retained as a supplementary identification strategy, not the primary design.

Two broader implications follow from this assessment for the design of the evaluation.

The first concerns what the absence of a feasible primary experimental design tells the evaluation about itself. The fact that randomisation cannot be deployed at BSCHAP is not a methodological inconvenience but a substantive finding: it reflects the ethical character of the intervention (an essential service, not an optional benefit), the operational reality of the catchment (a small, already-served population), and the retrospective character of the evaluation (a programme that exists, not one being designed). These features point toward a quasi-experimental identification strategy that constructs its counterfactual through statistical comparison with similar non-treated communities rather than through experimental allocation.

The second concerns the framing of the evaluation question itself. The ethical concerns about experimental designs are sharpest precisely because the intervention is known to be beneficial. Where the prior probability of programme effect is high — as is the case for primary healthcare access in a population with no alternative — withholding access for research purposes cannot be justified on the basis that the evidence generated would resolve genuine uncertainty about whether the programme works. This is one of the reasons the evaluation is framed in Section 1 as a magnitude-estimation rather

than hypothesis-testing exercise: the policy-relevant question is by how much BSCHAP improves outcomes, for whom, and at what cost, not whether it improves outcomes at all.

Section 3 has documented the methodological reasoning behind the rejection of experimental identification strategies as the primary approach to evaluating BSCHAP. Section 4 develops the quasi-experimental design that responds to the constraints surfaced here — a difference-in-differences specification combined with propensity score matching on a comparison group of geographically distant but ex-ante similar communities — and integrates randomised promotion as a supplementary identification strategy where uptake variation can be exploited.

Section 4: Quasi-Experimental Identification Strategy

4.1. From experimental rejection to quasi-experimental design

Section 3 concluded that experimental identification strategies are not feasible at BSCHAP. The absence of a credible randomisation moment is not an analytical inconvenience but a substantive feature of the intervention — an essential service provided to a community with no alternative, in a setting where the programme is already operational. The identification challenge for the evaluation is therefore to construct a credible counterfactual without randomisation, using the structure of the intervention and the comparability of treated and non-treated communities as the basis for causal inference.

The quasi-experimental toolkit offers three principal approaches to this problem. Regression discontinuity exploits a discontinuous treatment assignment rule — a threshold above which units are eligible and below which they are not — and identifies the treatment effect locally at the threshold. Difference-in-differences compares the change in outcomes over time between treated and non-treated units, identifying the treatment effect under the assumption that the two groups would have followed parallel trajectories in the absence of treatment. Propensity score matching constructs a synthetic comparison group by selecting non-treated units whose probability of treatment, given observed characteristics, most closely resembles that of treated units. Each method makes different identifying assumptions and is appropriate for different intervention structures.

This section assesses each method in turn against the BSCHAP context, then sets out the integrated strategy that the evaluation adopts. A central methodological choice that runs through the section is the level at which the analysis operates. The BSCHAP catchment contains approximately twelve treated villages and a comparable number of potential comparison villages; at this scale, village-level analysis is severely underpowered. The strategy adopted here moves the unit of analysis to the household level — preserving the village-level treatment assignment but conducting matching, regression, and inference at the unit of the household. This shift is the change that makes the design statistically credible. The justification, mechanics, and consequences of this choice are developed below.

4.2. Regression discontinuity (distance cutoff)

Regression discontinuity identifies a treatment effect by comparing units immediately above and below a discontinuous eligibility threshold. The identifying logic is that, near the cutoff, units on either side are similar in all respects except for treatment status, so the difference in their outcomes can be attributed to the treatment. For BSCHAP, the natural candidate cutoff is geographic distance from the clinic: villages within thirty kilometres of the project site would be defined as the treatment side of the cutoff, and villages between thirty and sixty kilometres would form the comparison just beyond the threshold. The

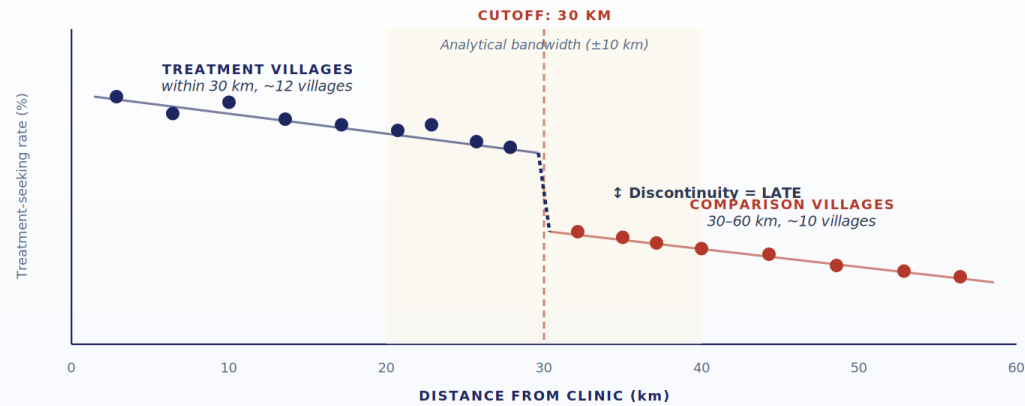
analytical bandwidth would be narrowed to villages within ten kilometres of the cutoff in either direction — that is, villages twenty to forty kilometres from the clinic — where comparability is maximised.

Figure 5 illustrates the design conceptually. Treatment-seeking rates are plotted against distance from the clinic; the discontinuity at thirty kilometres represents the local average treatment effect.

Regression Discontinuity (Distance Cutoff)

Villages on either side of a 30 km cutoff form the treatment and comparison groups; identification rests on observations near the threshold.

CONCEPTUAL DESIGN · TREATMENT-SEEKING RATE BY DISTANCE FROM CLINIC



WHY THIS DESIGN IS RULED OUT FOR BSCHAP

ARBITRARY CUTOFF

Clinical reach attenuates gradually with distance, not sharply at 30 km. A real RDD requires a discontinuous treatment rule — a defensible threshold below which households are categorically eligible and above which they are not.

BSCHAP has no such rule. Distance is a continuous covariate, not an eligibility threshold.

NUMEROSITY AT THE THRESHOLD

RDD identifies the effect only at the cutoff — using villages immediately on each side. The analytical bandwidth (20–40 km) leaves four to six villages on each side of the threshold.

The same numerosity problem that ruled out randomised designs reappears here, sharper.

MANIPULATION RISK

Households cannot manipulate where they live, but the implementing agency can redraw the "eligibility" rule. In a CSR context with a social-licence motive, pressure to include politically important villages on the treated side is real.

Cutoff integrity cannot be defended without an independent eligibility rule that does not exist.

VERDICT **RULED OUT.** No discontinuous eligibility rule exists, the bandwidth is severely underpowered, and cutoff integrity cannot be defended.

Figure 5. Regression Discontinuity (Distance Cutoff) — conceptual design and reasons for rejection.

Source: Author's design, following the RDD framework in Lee and Lemieux (2010).

The design is rejected on three connected grounds. The first is that the proposed cutoff is methodologically arbitrary. Regression discontinuity requires a discontinuous eligibility rule — a sharp categorical boundary between treated and untreated populations created by a programmatic or administrative decision. BSCHAP has no such rule. Clinical reach attenuates gradually with distance: a household five kilometres from the clinic is more likely to attend than a household twenty kilometres away, but the relationship is continuous, not discontinuous. The thirty-kilometre cutoff would be imposed by the evaluation rather than reflecting any feature of the intervention itself. Without an intrinsic discontinuity in treatment assignment, the design's identifying assumption — that units just above and below the cutoff are equivalent except for treatment — has no basis.

The second ground is statistical. Regression discontinuity identifies the effect only at the threshold, using observations within the analytical bandwidth. Narrowing the bandwidth to twenty to forty kilometres from the clinic leaves approximately four to six villages on each side of the cutoff. Whatever the merits of the design in principle, with this many units the design has no statistical power to detect effects of any plausible magnitude. The same numerosity constraint that ruled out randomised village-level designs in Section 3 reappears here in sharper form, because RDD restricts the analytical sample further than DiD or matching.

The third ground concerns the integrity of the cutoff itself. In a programmatic context, manipulation of the eligibility rule is a known threat to RDD identification. Communities cannot manipulate where they live, but the implementing agency can — and in a CSR context with a social-licence motive — has incentive to redraw the operational catchment in ways that include politically important villages. The cutoff is not protected from manipulation by an independent administrative rule. Without that protection, the design's credibility depends on a kind of institutional discipline that cannot be assumed in practice. Regression discontinuity is therefore ruled out as a primary identification strategy.

4.3 Difference-in-differences

Difference-in-differences identifies a treatment effect by comparing the change in outcomes over time between treated and comparison units. The first difference — within each unit, over time — controls for time-invariant characteristics that distinguish the two groups. The second difference — between the two groups' changes — controls for time trends that affect both groups equally. The residual is interpreted as the causal effect of treatment, under the identifying assumption that, in the absence of the intervention, the two groups would have followed parallel trajectories. This assumption — known as the parallel trends assumption — is the binding constraint on the credibility of any DiD estimate, and the evaluation design must address it explicitly.

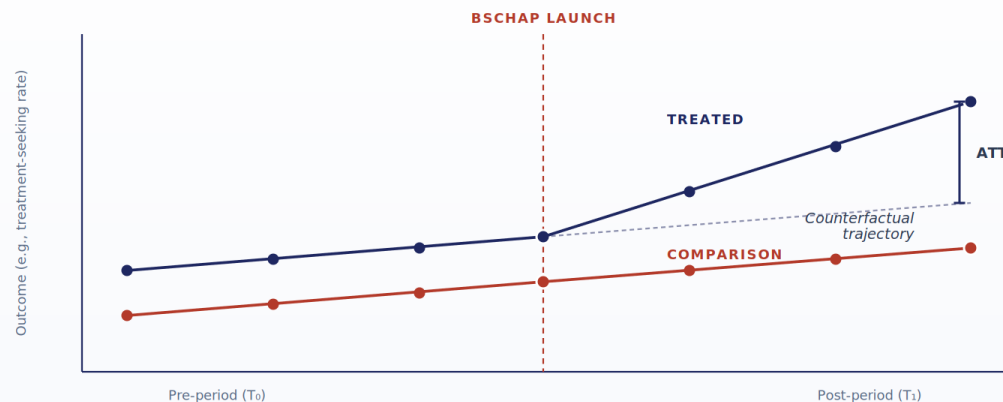
Figure 6 sets out the design as applied to BSCHAP.

FIGURE 6 · QUASI-EXPERIMENTAL DESIGN — OPTION B

Difference-in-Differences

Compares change in outcomes over time in treated communities to change in geographically distant but similar comparison communities.

CONCEPTUAL DESIGN · OUTCOME TRAJECTORIES BEFORE AND AFTER BSCHAP LAUNCH



DIFFERENCE-IN-DIFFERENCES ESTIMATOR

$$ATT = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre})$$

where:

$\bar{Y}_{T,t}$ = mean outcome among treated communities at time t

$\bar{Y}_{C,t}$ = mean outcome among comparison communities at time t

Interpretation

The first difference removes time-invariant differences between groups; the second difference removes shared time trends.

PARALLEL TRENDS ASSUMPTION

In the absence of BSCHAP, treatment and comparison communities would have followed parallel outcome trajectories.

Not the same as similarity in levels; tested by examining outcome trends for ≥ 2 pre-treatment periods.

DESIGN SPECIFICATION

TREATED COMMUNITIES

~12 villages within 30 km of project site
Pokot & Tugen pastoralist households
Population: 3,000-5,000
Receive BSCHAP services from launch ($T = T_0$)

Outcomes measured at household level —
not aggregated to village means

COMPARISON COMMUNITIES

~10-12 villages 50-80 km from project site
Geographically distant but ex-ante similar:
same livelihood, ethnicity, agro-ecological zone
No BSCHAP access; no clinic substitute

Not excluded from future programmes —
simply outside current catchment

DATA STRUCTURE

Household-panel survey at T_0 (baseline)
and T_1 (2-3 years post-launch)
~1,200 households per arm at each wave
Cluster-robust standard errors at village level

Sample large enough to detect
10-15 percentage-point effects

VERDICT **PRIMARY IDENTIFICATION STRATEGY.** Ethically clean, methodologically defensible, and adequately powered when implemented at household level.

Figure 6. Difference-in-Differences design — treated and comparison communities, outcome trajectories, and the ATT estimator.

Source: Author's design, following the DiD framework in Angrist and Pischke (2009).

Two features of the BSCHAP context make DiD particularly suitable as the foundation of the identification strategy. The first is that the comparison group exists naturally rather than being engineered by the evaluation. Communities fifty to eighty kilometres from the project site share the same livelihood system, ethnicity, agro-ecological zone, and county-level health policy environment as the treated communities, but lie outside the practical clinic catchment because of distance. They are not deliberately excluded from BSCHAP; they simply do not access the clinic because it is too far away. This means the comparison group can be assembled without raising the ethical concerns that defeated the experimental designs in Section 3.

The second is that the BSCHAP intervention has a clean treatment timing: the clinic opened on a defined date, allowing a sharp pre-versus-post distinction at the household level. Pre-programme data are available from clinic records covering the years immediately before community access began, and from routine demographic and health surveys conducted by the county health department. These pre-treatment data points are essential for testing the parallel trends assumption, which is what DiD ultimately rests on.

Testing parallel trends

It is worth being precise about what the parallel trends assumption requires and what it does not. The assumption is not that the two groups are similar in levels at baseline — it is entirely consistent with DiD for treated and comparison communities to have different starting points, as long as their trajectories would have moved in parallel. What is required is that the change over time in the comparison group is a credible counterfactual for the change that would have occurred in the treated group in the absence of treatment. The standard empirical test for this assumption examines outcome trends in the two groups during the pre-treatment period: if the trends are parallel before the intervention begins, the assumption that they would have remained parallel afterwards becomes credible.

This evaluation tests parallel trends using two pre-treatment data points — clinic records from the two years preceding community access for the implementing agency's catchment, and county-level health survey records for the broader comparison area. Where the pre-treatment trends in any indicator diverge between the two groups, that indicator is flagged for additional sensitivity analysis and is interpreted with appropriate caution in the results section.

4.4 Propensity score matching

Propensity score matching addresses a different identification problem from DiD: selection on observables. Even when two groups exist in the same setting and could in principle be compared, they may differ on observed characteristics — wealth, education, household composition, livelihood type —

that themselves drive outcomes. A naïve comparison would attribute to the treatment the part of the outcome difference that actually reflects these pre-existing differences. Matching addresses this by selecting, for each treated unit, one or more non-treated units that resemble it most closely on observed characteristics, and using these matched non-treated units as the comparison.

Direct matching on each observable becomes infeasible when the number of covariates is large, because exact matches become rare. Propensity score matching solves this by collapsing all observables into a single dimension — the propensity score, defined as the predicted probability that a unit receives treatment given its observed characteristics. Units with similar propensity scores are similar in their observable profile and can be treated as exchangeable for analytical purposes. The technique was formalised by Rosenbaum and Rubin in 1983 and has since become the standard tool for handling selection on observables in observational data.

The household-level reframing

This is the methodological choice that distinguishes the BSCHAP evaluation design from the village-level designs more commonly proposed in similar settings. The number of treated villages is approximately twelve; the number of potential comparison villages in the broader region is thirty or so. Matching twelve villages to twelve villages produces an analytical sample of twenty-four units — too few to support credible inference about anything but enormous effect sizes. Matching at the household level transforms the analytical sample dramatically: approximately 1,200 households in the treated catchment can be matched to approximately 1,200 households in the comparison pool, drawn from a wider universe of 3,500 candidates. The treatment indicator remains at village level (every household in a treated village is exposed to BSCHAP access; no household in a comparison village is), but the matching and the analysis operate at household level.

This reframing has three substantial implications for the design. First, the analytical sample size becomes adequate to detect treatment effects of plausible magnitudes at conventional confidence levels — a point developed in the power calculations in Section 4.6 below. Second, the heterogeneity analysis committed to in Section 1 — across women of reproductive age, children under five, pastoralist and agro-pastoralist sub-groups, and households at varying distances from the clinic — becomes actually feasible, because each sub-group contains enough households to support inference. Third, the matching can be done on a richer set of covariates than would be possible at village level, where many household-level characteristics aggregate uninformatively to village means.

The cost of household-level analysis under village-level treatment assignment is that observations are not independent. Households in the same village share environmental conditions, social networks, and access to the same clinic, and their outcomes are correlated. Standard errors that ignore this dependence will

understate the uncertainty around the treatment effect estimate. The evaluation handles this through cluster-robust variance estimators, with clustering at the village level. This is the standard solution in the literature and is well-supported in current statistical software.

Covariates and balance

Table 4 sets out the covariates on which propensity scores are estimated, the type of each, and the role each plays in the matching. The selection is guided by two principles. First, the covariate set must include any variable plausibly correlated with both treatment status (which depends on village location) and outcomes (which depend on household characteristics). Second, all covariates must be measured at baseline — that is, before any treatment effect could have occurred — so that matching on them does not absorb part of the effect we are trying to estimate.

Table 4. Covariates used in propensity score estimation

Covariate	Type	Role in matching
Household size	Continuous	Captures denominator for per-capita expenditure and health-use measures.
Female head of household	Binary	Demographic factor associated with health-seeking behaviour.
Mean education of adult members	Continuous	Proxy for health literacy and capacity to navigate formal care.
Livelihood category (pastoralist / agro-pastoralist / other)	Categorical	Determines mobility, resource patterns, and exposure to climate stress.
Household asset index	Continuous	Proxy for wealth — wealthier households face different access constraints.
Distance to nearest government health facility	Continuous	Controls for baseline access to alternative care, common to both arms.
Distance to nearest road and market	Continuous	Captures structural connectivity not collapsed into clinic distance.
Baseline treatment-seeking rate	Continuous	Pre-programme outcome itself, capturing latent health-care orientation.
Ethnic composition of village	Categorical	Linked to cultural orientation toward formal versus traditional care.

After matching, the evaluation tests covariate balance between the matched treated and matched comparison samples. Standardised differences of less than 0.1 standard deviations on each covariate are taken as evidence of adequate balance; any covariate failing this threshold is included as a control variable in the DiD regression as a robustness measure.

4.5. The integrated PSM-DiD strategy

Neither propensity score matching nor difference-in-differences alone is sufficient for credible identification in the BSCHAP setting. PSM addresses selection on observables but cannot address selection on unobservables — characteristics not measured in the survey that may simultaneously affect treatment exposure and outcomes. DiD addresses time-invariant unobservables through the within-unit comparison but does not address baseline imbalance on observables. Combining the two methods — PSM-DiD — handles both threats: matching equalises the two arms on observed covariates at baseline, and the subsequent DiD analysis differences out any residual time-invariant differences whether observed or not.

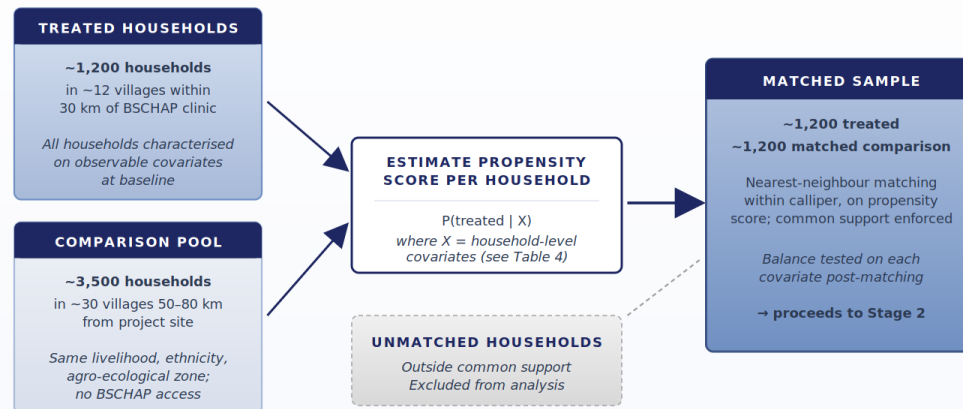
Figure 7 sets out the integrated strategy.

FIGURE 7 · RECOMMENDED IDENTIFICATION STRATEGY

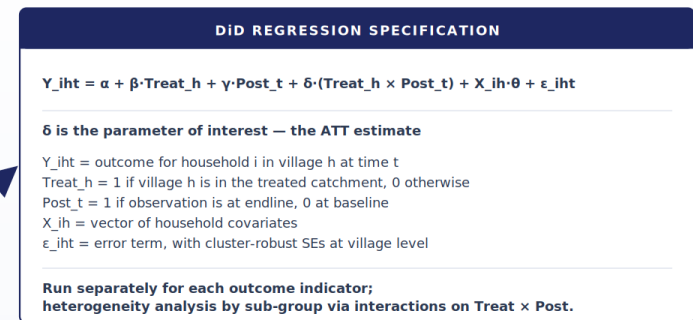
Propensity Score Matching with Difference-in-Differences (PSM-DiD)

Two-stage strategy: PSM constructs a matched comparison sample at household level; DiD then estimates the treatment effect on the matched sample over time.

STAGE 1 — PROPENSITY SCORE MATCHING (HOUSEHOLD LEVEL)



STAGE 2 — DIFFERENCE-IN-DIFFERENCES ON THE MATCHED SAMPLE



WHY THIS DESIGN

PSM ADDRESSES SELECTION ON OBSERVABLES

Households in treated and comparison communities differ on observable characteristics that may also drive outcomes. Matching on the propensity score balances the two arms on those covariates and restricts analysis to the region of common support.

Limitation: cannot match on unobservables.

DID ADDRESSES TIME-INVARIANT UNOBSERVABLES

After matching, residual differences between arms on unobserved characteristics may remain. DiD handles these by differencing them out — any time-invariant confounder is removed by the within-unit comparison over time.

Limitation: time-varying unobservables remain a threat.

HOUSEHOLD-LEVEL ANALYSIS RESTORES POWER

With ~1,200 households per arm at each wave, the sample is large enough to detect 10–15 percentage-point treatment effects at conventional confidence levels — including heterogeneity across sub-groups.

Village-level matching would have been underpowered.

VERDICT **RECOMMENDED PRIMARY DESIGN.** Combined PSM-DiD at household level addresses selection on observables, time-invariant unobservables, and the numerosity constraint that ruled out other designs.

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Figure 7. PSM-DiD identification strategy — two-stage design integrating matching at household level with difference-in-differences analysis.

Source: Author's design, following the PSM-DiD approach in Heckman, Ichimura and Todd (1997).

The DiD regression on the matched sample is the central estimator of the evaluation. The treatment effect parameter — the coefficient on the interaction between treatment-arm assignment and post-treatment indicator — represents the average treatment effect on the treated (ATT), the magnitude of effect for households that actually received BSCHAP access. Where heterogeneity analysis is required, the same specification is extended with triple interactions involving the sub-group of interest, allowing separate effect estimates for women of reproductive age, children under five, pastoralist versus agro-pastoralist households, and households at varying distances from the clinic.

4.6 Sampling, data collection, and power

The sampling strategy follows the design specification developed above. In the treatment arm, all households in the approximately twelve catchment villages are enumerated through a household listing exercise at baseline; the analytical sample is the full universe of treated households (estimated at 1,200) rather than a sample, because the catchment is small enough that full enumeration is operationally and financially feasible. In the comparison arm, the pool of approximately thirty potential comparison villages is reduced first by exclusion of villages that fail eligibility criteria (presence of a competing health facility, fundamentally different livelihood structure, security inaccessibility), then by full household enumeration in the remaining ten to twelve villages. The propensity score matching then operates on the combined treated and comparison household lists.

Data collection uses a structured household survey administered at two waves — baseline before BSCHAP services begin in any new community recruited for the evaluation (or, for the existing catchment, baseline data reconstructed from clinic records and retrospective household recall) and endline approximately two to three years later. The survey instrument captures the outcome indicators specified in Section 2, the household-level covariates used for matching, and the demographic and contextual variables needed for heterogeneity analysis. Clinic register data provide a triangulation source for service-utilisation indicators. The detailed survey instrument is included in Annex A of this design document.

Power and minimum detectable effects

Power calculations for the PSM-DiD design on a matched household sample produce the minimum detectable effects shown in Table 5. The calculations assume 1,200 households per arm at each wave, an intra-cluster correlation of 0.05 (typical for health indicators in this kind of setting), a significance level of 5%, and a power target of 80%.

Table 5. Minimum detectable effects under the recommended design

Outcome indicator	Baseline rate (assumed)	Minimum detectable effect at 80% power
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Treatment-seeking rate	30%	+10 percentage points
ANC completion (4+ visits)	25%	+11 percentage points
Full childhood immunisation	40%	+11 percentage points
Skilled birth attendance	20%	+10 percentage points
Catastrophic health expenditure	35% of households	–12 percentage points

The minimum detectable effects in Table 5 are in the range of ten to twelve percentage points for the primary outcome indicators. This is the threshold below which the design cannot reliably distinguish a true programme effect from random variation. Effect sizes of this magnitude are plausible for BSCHAP, given the absence of any prior primary care provision in the catchment and the strong literature on the impact of facility-based health service introduction in similar settings. The design is therefore adequately powered for its intended purpose, on the condition that it is implemented at household level. A village-level analysis with the same number of villages — twelve treated and ten to twelve comparison — would have a minimum detectable effect of approximately thirty to forty percentage points, which is far larger than any plausible programme impact and would therefore have failed to detect effects of policy interest with high probability. This is the binding methodological reason for the household-level reframing.

4.7 Limitations and how the design addresses them

No quasi-experimental design eliminates all threats to identification. Three limitations of the PSM-DiD strategy warrant explicit acknowledgement, along with the robustness checks the design includes to mitigate each.

The first limitation is that matching can only balance the two arms on observed characteristics. Unobserved differences between treated and comparison households — community leadership quality, latent attitudes toward formal care, prior relationships with the implementing agency — remain a source of bias. The DiD layer addresses this in part, by differencing out any unobservable that is time-invariant. What remains is the threat of time-varying unobservables that affect the two arms differently. The evaluation addresses this through two robustness checks: a placebo test using a falsified treatment date (testing whether the apparent treatment effect appears spuriously in the pre-treatment period), and a sensitivity analysis to selection on unobservables using the Rosenbaum bounds approach.

The second limitation is the parallel trends assumption itself. If treated and comparison communities are on diverging trajectories for reasons unrelated to BSCHAP — for example, if the broader project's economic activity is changing wealth or migration patterns in treated villages but not in distant

comparison villages — the DiD estimate captures this divergence as if it were a treatment effect. The evaluation tests parallel trends using two pre-treatment data points and includes an event-study specification as a robustness check, which allows year-by-year examination of the divergence between the two arms.

The third limitation concerns the randomised promotion strategy retained from Section 3 as a supplementary identification approach. Where promotion intensity can be varied across villages — through differential CHV deployment, transport subsidy schemes, or community outreach intensity — the resulting variation in uptake provides an additional source of identification through instrumental variables estimation. The IV estimate of the LATE serves as a triangulation check on the PSM-DiD estimate: if the two methods, which rely on different identifying assumptions, produce similar effect estimates, the credibility of the causal claim is substantially strengthened. The integration of the randomised promotion supplementary analysis into the overall analysis plan is developed in Section 5.

The recommended identification strategy combines propensity score matching with difference-in-differences at household level. PSM addresses selection on observables; DiD addresses time-invariant unobservables; household-level analysis restores the statistical power that village-level analysis cannot provide. The strategy is adequately powered, ethically defensible, and methodologically appropriate to the structure of BSCHAP as an established service-delivery intervention in a small but well-defined catchment.

Section 5: Mixed-Methods Strategy — Qualitative Approach

5.1 Why qualitative methods are essential, not optional

The quantitative identification strategy developed in Section 4 — propensity score matching combined with difference-in-differences at household level — is designed to produce credible estimates of the average treatment effect of BSCHAP access on the outcome indicators specified in Section 2. It is designed to answer one specific kind of question: whether and by how much BSCHAP changed those outcomes, on average, for households in the treated catchment. It is not designed to answer questions about how the programme produced those changes, why some households were affected more than others, what beneficiaries experienced when they accessed the clinic, or what consequences the programme had that lie outside the pre-specified indicator set.

These are not minor or supplementary questions. They are the questions that determine whether the evaluation's findings can actually be used to improve the programme, replicate it elsewhere, or defend it to the implementing agency's senior leadership. A point estimate of a ten-percentage-point increase in treatment-seeking behaviour is methodologically defensible but operationally limited: it tells the CSR director that the programme worked at some scale, but not why some villages responded and others did not, not which features of the service mix were most consequential, not which barriers remained binding for non-users, and not whether the programme produced effects on community-implementing-agency relations that affect the social licence of the broader project. Each of these matters for decision-making, and each requires qualitative evidence.

There is a more specific reason why qualitative methods are essential to this particular evaluation. The theory of change developed in Section 2 identifies five layers of assumptions on which the causal claim depends — management, implementation, behavioural and uptake, development, and sustainability. The PSM-DiD strategy in Section 4 produces an effect estimate conditional on these assumptions holding; it does not test the assumptions themselves. Two of the five layers — implementation assumptions about access disruption and the operation of referral linkages, and behavioural-uptake assumptions about community awareness of entitlement, trust in the service, and the absence of prohibitive transport costs — are particularly amenable to qualitative investigation. Where the qualitative strand finds these assumptions broadly upheld, the credibility of the causal claim is strengthened. Where it finds them violated, the quantitative estimate must be interpreted in the light of that violation.

The qualitative strand is therefore not a separate evaluation conducted in parallel to the quantitative one. It is part of a single mixed-methods evaluation design in which the two strands address different facets of the same impact criterion, integrate through deliberate analytical procedures, and produce a body of evidence neither could produce alone. Figure 8 on the following page sets out the integration architecture.

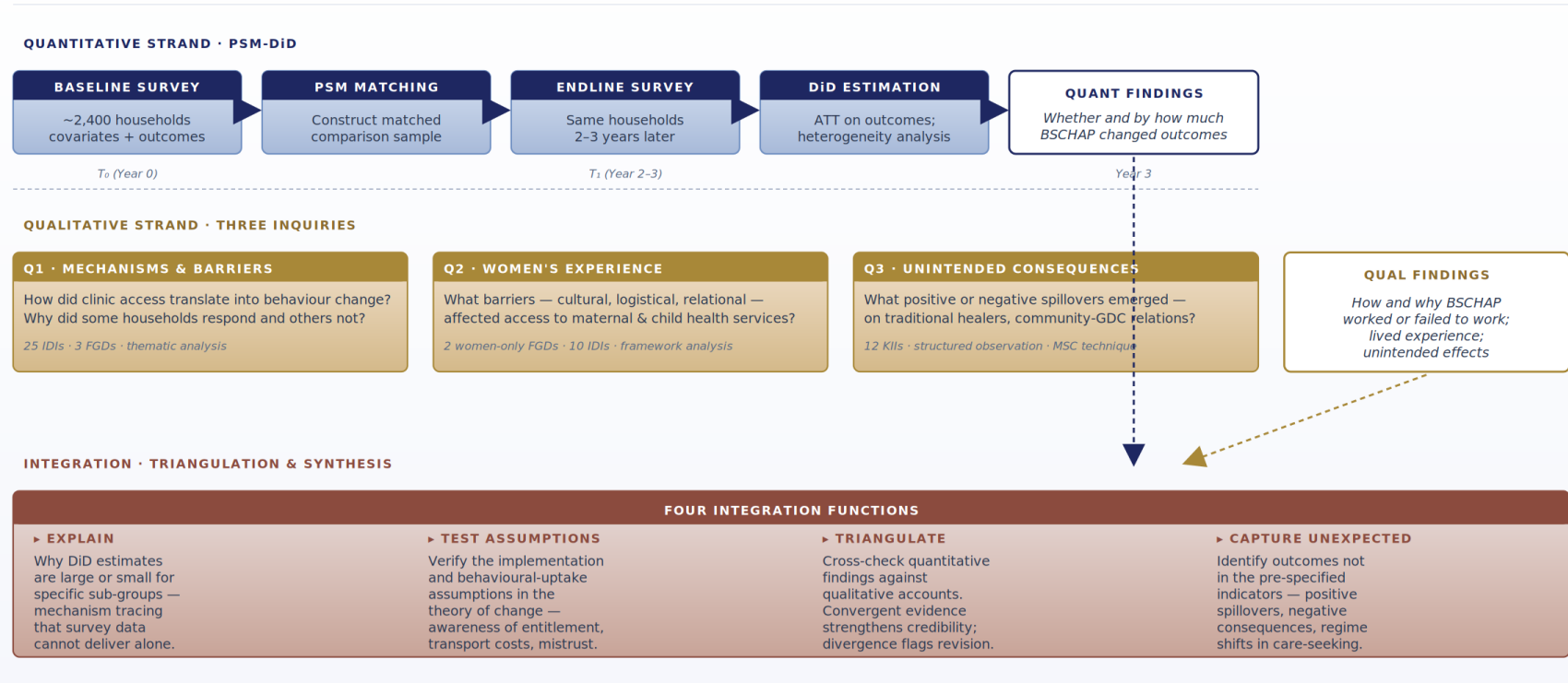
5.2 The mixed-methods integration architecture

Figure 8 shows how the qualitative and quantitative strands are sequenced over time and integrated at the analytical layer. Both strands follow the same project timeline: baseline (T_0), implementation period, and endline (T_1) approximately two to three years after BSCHAP launch. The quantitative strand uses both waves to construct the DiD estimator; the qualitative strand uses the endline period for its primary data collection, with a smaller scoping component at baseline. The integration occurs at the analytical layer, where qualitative findings are used to explain, test, triangulate, and complement the quantitative estimates.

FIGURE 8 · MIXED-METHODS INTEGRATION

How the Qualitative Strand Connects to the Quantitative Design

Qualitative work is sequenced and integrated with the PSM-DiD analysis to address what quantitative methods alone cannot answer.



DESIGN PRINCIPLE

Quantitative answers **whether and by how much**; qualitative answers **how and why**. Neither alone produces decision-useful evidence in a setting of known-beneficial treatment.

BSCHAP IMPACT EVALUATION DESIGN · SECTION 5 · EVALUATION DESIGNER: ERIC KIZAMBO KACHILA

Figure 8. Mixed-methods integration architecture — how the qualitative strand connects to the PSM-DiD design over time and analytically.

Source: Author's design, drawing on Creswell and Plano Clark's convergent parallel mixed-methods framework (2018).

Four integration functions structure the relationship between the two strands. The first is explanation: qualitative findings on mechanism and barrier dynamics explain why the quantitative point estimates take the values they do, particularly where heterogeneity analysis surfaces different effect sizes for different sub-groups. The second is assumption testing: qualitative inquiry verifies whether the implementation and behavioural-uptake assumptions in the theory of change actually held during the evaluation period. The third is triangulation: where the two strands converge on the same finding, the credibility of the evaluation's claims is strengthened; where they diverge, the divergence is itself a finding that demands explanation. The fourth is the capture of unexpected outcomes that fall outside the pre-specified indicator set — positive spillovers, negative externalities, or shifts in care-seeking regimes that the quantitative design was not built to detect.

This architecture reflects a specific design choice. The qualitative strand is positioned as analytically equal to but methodologically distinct from the quantitative strand, with neither dominating the other. This is the convergent parallel mixed-methods design in Creswell and Plano Clark's typology — distinct strands collecting and analysing their own data in their own ways, then meeting at the analytical layer for joint interpretation. The alternative — a sequential explanatory design, where qualitative work follows quantitative results to explain them — would treat qualitative methods as derivative of quantitative findings and would miss the assumption-testing and unexpected-outcomes functions that require qualitative inquiry to be planned independently. The choice of convergent parallel is therefore methodologically deliberate rather than conventional.

5.3 The three qualitative evaluation questions

Three evaluation questions structure the qualitative strand. Each is framed to address a specific gap left by quantitative methods alone, and each contributes a different function to the integrated evidence base. Figure 9 below presents the full evaluation matrix; the discussion that follows the matrix unpacks the rationale for each question and the methodological choices that follow from it.

FIGURE 9 · QUALITATIVE EVALUATION MATRIX

Three Inquiries · Data Sources · Methods

Each impact evaluation question is paired with the data sources that can credibly answer it and the methods used to collect and analyse those data.

EVALUATION QUESTION	SOURCES OF DATA	COLLECTION & ANALYSIS METHODS
Q1 · MECHANISMS & BARRIERS Through what mechanisms did access to BSCHAP change health-seeking behaviour, and why did some households change while others did not? SUB-QUESTIONS 1a. Which clinic services or interactions most influenced decisions to seek care? 1b. What barriers — awareness, transport, cultural norms, mistrust — remained? 1c. How did CHVs and community leaders influence service uptake?	▶ BSCHAP users (direct beneficiaries) ▶ Community members who did NOT use services despite availability ▶ Community Health Volunteers ▶ Village elders & community leaders ▶ BSCHAP clinical staff (clinical officer, nurse) ▶ Clinic attendance registers (for triangulation only)	In-depth interviews (IDIs) 25 interviews · users & non-users · stratified by gender and age · 60 min each · Pokot/Tugen with interpreter Focus group discussions (FGDs) 3 FGDs · women using maternal services; household heads; CHVs and leaders · 10 participants each · 90 min Analysis Thematic content analysis with systematic coding; user vs non-user comparison; triangulation with attendance data. <i>Timeframe: 6 weeks.</i>
Q2 · WOMEN'S EXPERIENCE How did women in target communities experience BSCHAP services, and what barriers — cultural, logistical, relational — affected access to maternal and child health services? SUB-QUESTIONS 2a. Did women feel safe and respected at the clinic? 2b. What role did husbands/decision-makers play? 2c. Which specific services were most accessible?	▶ Women of reproductive age (users and non-users of maternal services) ▶ BSCHAP nurse — maternal services lead ▶ Traditional birth attendants (TBAs) ▶ Male household heads ▶ County health office records on maternal health indicators in Baringo	Women-only FGDs 2 FGDs · users and non-users · female facilitator required · 10 participants each · 90 min · private settings IDIs with TBAs & male household heads 6 IDIs with TBAs on traditional birth practices and BSCHAP uptake; 3 KIIs with male heads on decision-making Analysis Gender-sensitive thematic analysis; framework analysis comparing user vs non-user experience. <i>Timeframe: 4 weeks.</i>
Q3 · UNINTENDED CONSEQUENCES What unintended positive or negative consequences did BSCHAP produce — on community-GDC relations, traditional healing, or intra-household decisions? SUB-QUESTIONS 3a. Did BSCHAP improve or damage community perceptions of GDC and its operations? 3b. How did access affect traditional healers? 3c. Were there inequitable access patterns within or between communities?	▶ Community elders & opinion leaders ▶ Traditional healers & herbalists active in target communities ▶ GDC Community Liaison & CSR Manager ▶ BSCHAP clinic staff ▶ Comparison village representatives (to assess spillover) ▶ GDC engagement records & clinic logs	Key informant interviews (KIIs) 12 KIIs · elders, GDC liaison, traditional healers, comparison-village reps · 60 min each Structured observation 5 days at the clinic · service quality, patient interactions, equity in access across villages Analysis · Most Significant Change technique Case study synthesis; MSC stories identify the most consequential planned and unplanned changes. <i>Timeframe: 4 weeks.</i>

BSCHAP IMPACT EVALUATION DESIGN · SECTION 5 · EVALUATION DESIGNER: ERIC KIZAMBO KACHILA

Figure 9. Qualitative evaluation matrix — three inquiries, their data sources, and the collection and analysis methods that operationalise them.

Source: Author's design, applying the standard evaluation matrix structure used in OECD-DAC and 3ie evaluation guidance.

Q1 — Mechanisms and barriers

The first question asks through what mechanisms BSCHAP changed health-seeking behaviour, and why some households changed while others did not. This is the core mechanism-tracing inquiry of the qualitative strand. Where the quantitative analysis will produce an average treatment effect on, for example, formal treatment-seeking rates, this question explains the variance underlying that average — which features of the service mix mattered, which barriers remained binding, and which informal intermediaries (community health volunteers, village elders, peer networks) shaped uptake.

Three sub-questions structure the inquiry. The first asks which specific clinic services or interactions most influenced community members' decisions to seek care. This addresses the operational specificity that policy users need: not whether the clinic worked, but which components of the service offering did the work. The second asks what barriers — awareness of entitlement, transport costs, cultural norms, mistrust of formal providers — remained even after clinic access was available. The framing follows the revised assumption structure in Section 2: where the original logic model treated barriers vaguely, the revised framework names specific behavioural-uptake assumptions whose validity this question is designed to test. The third asks how community health volunteers and community leaders influenced service uptake, addressing the role of informal intermediaries that quantitative survey data cannot capture.

Data sources for this question span both users and non-users of BSCHAP services. The inclusion of non-users is methodologically critical: a question about why some households did not change behaviour cannot be answered by speaking only to those who did. Twenty-five in-depth interviews, stratified by gender and age and conducted in Pokot and Tugen with skilled interpreter support, form the primary data source. Three focus group discussions — one with women using maternal services, one with household heads, and one with community health volunteers and leaders — provide group-level dynamics that individual interviews cannot surface. The clinic attendance registers are used not as a primary data source but as a triangulation reference for the patterns described by interview participants. Thematic content analysis, with systematic coding and explicit comparison across user and non-user groups, structures the analytical work.

Q2 — Women's experience

The second question asks how women in the target communities experienced BSCHAP services, and what cultural, logistical, and relational barriers affected their access to maternal and child health services. This is a gendered access inquiry, and its inclusion reflects both the substantive importance of maternal and child health in BSCHAP's outcome indicator set and the methodological reality that gendered access

barriers in pastoralist communities are largely invisible to household-level surveys conducted by mixed-gender enumerators.

Three sub-questions guide the work. The first asks whether women felt safe and respected when accessing the clinic and whether services were culturally appropriate. This addresses a dimension of service quality that does not appear in any of the pre-specified outcome indicators but materially shapes whether women return, recommend the clinic to others, or continue to engage with formal maternal services after a single experience. The second asks what role husbands and household decision-makers played in women's ability to access services, surfacing the intra-household dynamics that mediate the relationship between formal entitlement and actual access. The third asks which specific maternal and child services — antenatal care, delivery support, child immunisation — women found most or least accessible, providing the service-specific detail that informs targeted programme adjustment.

The methodological approach is shaped by the substantive content. Two women-only focus group discussions, one with users and one with non-users of maternal services, are conducted by a female facilitator in private settings to enable safe disclosure of experiences that participants may be reluctant to share in mixed company. Six in-depth interviews with traditional birth attendants explore the relationship between traditional birth practices and BSCHAP uptake, surfacing both substitution effects and complementary dynamics. Three key informant interviews with male household heads investigate the decision-making structures that determine women's access to formal care. Gender-sensitive thematic analysis, supplemented by framework analysis comparing the experiences of women who did and did not access maternal services, structures the analytical work.

Q3 — Unintended consequences

The third question asks what unintended positive or negative consequences BSCHAP produced for the broader community, including effects on community-implementing-agency relations, the role of traditional healers, and intra-household health decision dynamics. This is a boundary inquiry: it identifies outcomes that fall outside the pre-specified indicator set and that the quantitative design was not built to detect.

This question matters for three reasons specific to the BSCHAP context. The first is that the implementing agency's social licence to operate the broader geothermal project depends materially on community relations. If BSCHAP improves or damages community perceptions of the implementing agency, that is a consequential outcome regardless of whether it appears in the health indicator set. The second is that introducing a free formal health facility into a context where traditional healers are active providers may shift the role those healers play — toward complementarity, displacement, or some hybrid — and that shift has both substantive and ethical implications that warrant explicit investigation. The third is that any

intervention operating at village level produces equity effects within and between communities that household-level outcome indicators may aggregate over.

Twelve key informant interviews with community elders, traditional healers, herbalists, the implementing agency's community liaison officer and CSR manager, and representatives from comparison villages cover the principal actors. Structured observation at the clinic over five days assesses service delivery quality, patient interactions, and equity in access patterns across villages and sub-groups. Document review draws on the implementing agency's community engagement logs, the clinic's complaint register, and community meeting minutes referencing BSCHAP. The analytical work applies the Most Significant Change technique — a participatory approach in which stakeholders identify the changes they consider most consequential and the reasons for their selection — alongside case study analysis synthesising all qualitative data.

5.4 Methodological functions and integration

Table 6 summarises how each of the three evaluation questions contributes to the integrated mixed-methods evidence base. The table makes explicit the relationship between each qualitative inquiry and the quantitative analysis it supports.

Table 6. Methodological functions of the three qualitative inquiries

Inquiry	Methodological function	Primary integration with PSM-DiD
Q1 — Mechanisms and barriers	Mechanism tracing: explains how clinic access translated into behaviour change and why uptake varied across households.	Explains heterogeneity in DiD estimates; tests behavioural-uptake assumptions in the theory of change.
Q2 — Women's experience	Lived experience: surfaces gendered access barriers that household-level surveys cannot capture.	Provides interpretive context for sub-group treatment effects on maternal and child outcomes.
Q3 — Unintended consequences	Boundary identification: captures outcomes outside the pre-specified indicator set.	Surfaces effects the quantitative design was not built to detect; informs interpretation and policy recommendations.

The integration is operationalised through joint display analysis at the synthesis stage. For each evaluation question in the overall evaluation, the corresponding quantitative estimates and qualitative themes are presented side-by-side in a single analytical display, allowing the evaluator to identify points of convergence, divergence, and complementarity. Convergent findings — where the DiD estimate of effect

on a particular outcome aligns with the qualitative account of mechanism and barrier — strengthen the credibility of the causal claim. Divergent findings — where qualitative evidence contradicts the quantitative pattern, or vice versa — are not treated as problems to be resolved by giving one strand priority over the other. They are treated as substantive findings about the limits of either strand's reach, and reported as such in the final evaluation report.

5.5 Fieldwork sequencing

The sequencing of qualitative and quantitative fieldwork over the evaluation timeline matters both methodologically and operationally. Table 7 sets out the activity schedule and the integration points across the evaluation period.

Table 7. Sequencing of qualitative and quantitative fieldwork

Phase	Quantitative activity	Qualitative activity	Integration
Pre-baseline (months 1-2)	Sampling frame, instrument design	Scoping KIIs with CHVs, traditional healers, GDC liaison; refines survey item wording	Qualitative scoping informs quant instrument cultural appropriateness and translation choices.
Baseline (months 3-5)	Household survey administered in treatment and comparison communities	Light parallel qualitative: 5-6 IDIs to test instrument comprehension and capture pre-treatment narratives	Qual pretest validates quant baseline; pre-treatment stories anchor later before/after comparisons.
Implementation period (months 6-30)	No active data collection; clinic records monitored	No active data collection; CHV monthly reflection notes archived	Both strands quiet; data archived for later use.
Endline (months 30-36)	Endline household survey; PSM matching; DiD estimation; heterogeneity analysis	Full qualitative fieldwork: 25 IDIs (Q1), 2 women-only FGDs + 10 IDIs (Q2), 12 KIIs + 5 days observation (Q3)	Qual conducted concurrently with quant analysis to allow iteration. MSC stories triangulated against DiD point estimates.
Synthesis (months 36-38)	Final regression tables, robustness checks, sensitivity analyses	Coded thematic outputs, framework analysis tables, MSC narratives	Joint display analysis: quant estimates and qual themes presented side-by-side for each evaluation question.

Two design choices in the sequencing warrant brief explanation. The first is the inclusion of a small qualitative scoping component before the quantitative baseline. The purpose is not to generate findings but to inform the quantitative instrument design — specifically, the wording of survey items, the cultural appropriateness of indicator definitions, and the translation choices that determine whether the baseline survey captures what it is designed to capture. Without this scoping, the quantitative instrument risks measuring something other than what it intends to measure, and no amount of statistical sophistication at the analysis stage can recover from poor measurement at the data collection stage.

The second is the concentration of the main qualitative fieldwork at endline rather than throughout the evaluation period. This reflects an analytical judgement that the most valuable qualitative data — accounts of how BSCHAP changed lives, what barriers remained binding, what unintended consequences emerged — are most accurately collected when participants have something to reflect on, which requires the programme to have been operating for the full evaluation period. Continuous qualitative monitoring would have produced richer process documentation but at much higher cost and with less analytical leverage. The choice of endline concentration trades depth of process documentation for analytical focus on impact, which matches the evaluation's purpose.

5.6 Ethical considerations specific to the qualitative strand

The qualitative strand introduces ethical considerations that the quantitative strand does not. Five warrant explicit attention in this design.

The first is consent in low-literacy settings. Written informed consent is not appropriate for most participants in the catchment; verbal consent must be obtained in the participant's first language, with explicit acknowledgement that the participant can withdraw at any point without consequence and that withdrawal will not affect their access to BSCHAP services or to the implementing agency more broadly. The latter clause is critical: in a context where the evaluator and the implementing agency are perceived as connected, the risk that participants feel coerced into participation is real and must be actively mitigated through the consent protocol.

The second is confidentiality in small communities. Anonymity in the conventional sense is difficult to guarantee when a village has thirty households and the evaluation team interviews three of them. The protocol must therefore commit not to ascription of specific quotes to specific individuals in the final report, and to careful aggregation of findings to the village or sub-group level in any text where identifying detail might be inferred. The implementing agency does not receive raw interview transcripts; it receives the analytical synthesis only.

The third is the discussion of sensitive topics with women, particularly around maternal health, reproductive decisions, and intra-household power dynamics. Women-only focus group discussions are

conducted by a female facilitator in private settings; participants are explicitly told they need not answer any question they would prefer not to answer; referral resources are pre-arranged in case discussion surfaces unresolved health needs that should not be ignored.

The fourth is the involvement of traditional healers as a data source. The framing of the inquiry must not position traditional healing practices as obstacles to formal care; the design treats traditional healers as legitimate health-system actors whose role is itself the object of inquiry, not a problem to be diagnosed. The interview protocol is drafted with this stance explicitly in view.

The fifth is the management of findings that may be uncomfortable for the implementing agency. Where qualitative findings surface community grievances against the implementing agency, criticisms of the broader geothermal project, or accounts of negative experiences with BSCHAP staff, the protocol commits to reporting these findings transparently in the final evaluation. The evaluator's independence is the source of the evaluation's credibility; any compromise of that independence at the reporting stage would undermine the value of the entire exercise.

5.7 Limitations of the qualitative strand

Three limitations of the qualitative strand warrant explicit acknowledgement in the design.

The first is generalisability. Qualitative findings from twenty-five in-depth interviews and five focus group discussions are not statistically generalisable to the catchment population in the way that a household survey is. The qualitative strand can document patterns, identify mechanisms, and surface unexpected outcomes, but it cannot tell us what proportion of the catchment population experienced any given phenomenon. The quantitative strand carries the burden of population-level inference; the qualitative strand carries the burden of explanation and meaning. The evaluation's credibility depends on neither strand attempting to do the work of the other.

The second is the translation and interpretation chain. Most interviews and focus groups are conducted through an interpreter working between Pokot or Tugen and English, then transcribed and translated again for analytical coding. Each stage in the chain introduces interpretive choices that affect what the evaluator ultimately analyses. The design mitigates this through interpreter briefing on the evaluation's analytical priorities, back-translation of key quotes, and inclusion of original-language transcripts in the analytical record. Complete elimination of translation-related interpretive loss is not possible.

The third is the potential for desirability bias in interviews conducted under the auspices of an evaluation linked to the implementing agency. Participants may report more favourable experiences with BSCHAP than they actually had, particularly if they perceive the interviewer as connected to the agency. The design addresses this through explicit framing of the evaluator's independence at the start of each interview,

through the inclusion of non-users of BSCHAP services in the sample (who have less reason to manage their accounts for the implementing agency), and through triangulation of self-reported experiences against the structured observation data and the clinic complaint register. The bias is mitigated rather than eliminated.

The qualitative strand operationalises the principle that quantitative methods answer whether and by how much, while qualitative methods answer how and why. Neither alone produces decision-useful evidence in a setting where the treatment is known to be broadly beneficial; together, they produce the integrated mixed-methods evidence base that policy and management decisions actually need.

Section 6: Implementation Framework

6.1 Implementation philosophy

An impact evaluation is a research exercise only in part. It is also, and more substantially, an operational endeavour: it requires teams to be assembled, fieldwork to be coordinated across remote geographies, ethical approvals to be secured, data to be collected at quality, analytical work to be conducted to time and budget, and findings to be communicated to people whose decisions the evaluation is intended to inform. The design developed in Sections 1 through 5 is methodologically defensible only if the implementation arrangements set out here can credibly deliver it. The two halves of the document — design and implementation — are equally important for the document's standing as a piece of professional work.

Three principles shape the implementation arrangements proposed below. The first is internal leadership with external technical depth. The evaluation is led by the implementing agency's monitoring and evaluation function rather than outsourced wholesale to an external evaluator, but the technical work — PSM matching, DiD estimation, qualitative analysis — is conducted by short-term external specialists who bring methodological independence and analytical rigour that an internal team cannot reasonably hold across the full duration of an evaluation conducted infrequently. This hybrid model is operationally realistic for a CSR-funded evaluation of this scale, ethically defensible because external specialists conduct the analytical work that informs the causal claims, and financially efficient because the internal coordination cost is absorbed into existing M&E function rather than added to the evaluation budget.

The second is locally-recruited fieldwork. Enumeration and qualitative fieldwork are conducted by personnel recruited locally through the county-level M&E network, in collaboration with the county health department. This is not only a cost choice: locally recruited enumerators speak Pokot and Tugen fluently, understand the cultural conventions of household entry, can navigate the seasonal logistics of pastoralist communities, and earn community trust faster than externally imported teams. Pairing local field staff with external analytical specialists produces a stronger evaluation than either purely external or purely internal approaches deliver.

The third is integration of qualitative and quantitative strands at the personnel level. The same internal M&E lead coordinates both strands and is responsible for the integrated synthesis at endline. The qualitative researcher and external specialist communicate throughout the analytical period rather than working in parallel silos and meeting only at the joint-display analysis stage. This is operationally consequential: the convergent parallel mixed-methods design developed in Section 5 depends on iterative cross-strand learning, which depends in turn on personnel structures that enable that learning.

6.2 Governance

The evaluation's governance structure assigns clear decision-making authority, separates analytical work from programme management, and provides escalation pathways for issues that arise during implementation. Four governance roles are specified.

The CSR director of the implementing agency holds ultimate responsibility as the evaluation's sponsor. The director approves the inception report, signs off on the budget envelope, chairs the project steering committee, and receives the final evaluation report. The director does not engage with the analytical conduct of the evaluation; that separation is essential to the evaluation's credibility and to the social licence of the broader project to which BSCHAP belongs.

The M&E lead within the implementing agency serves as the evaluation manager. The M&E lead translates the design into day-to-day operational decisions, coordinates the contracting of external specialists, manages the field operations, oversees the quality assurance protocols described in Section 6.7, and prepares the deliverables for steering committee review. The M&E lead is the locus of operational accountability for the evaluation.

A small steering committee, meeting at the inception, mid-point, and synthesis milestones, provides oversight without operational involvement. The committee includes the CSR director, the M&E lead, a representative of the county health department, and an independent member with evaluation expertise (e.g. drawn from a Kenyan university research office or a development agency M&E function in Nairobi). The independent member is the structural safeguard against capture of the evaluation by the implementing agency's reputational interests.

The external specialists — quantitative and qualitative — sit outside the steering committee structure and report functionally to the M&E lead. Their analytical decisions are recorded in a methodological log that is appended to the final evaluation report as an annex, ensuring transparency about the analytical choices made and the reasoning behind them. The specialists' analytical independence is contractually protected: their contracts specify that the final analytical findings, whatever they are, will be presented to the steering committee without alteration by the implementing agency.

6.3 Team composition

Table 8 sets out the team required to deliver the evaluation, the responsibility of each role, and the time commitment associated with each.

Table 8. Evaluation team composition and time commitments

Role	Responsibility	Time commitment	Source
M&E lead (internal)	Overall coordination; stakeholder liaison; quality assurance; report drafting in synthesis phase.	~40 days over 38 months	Implementing agency M&E function
External specialist	PSM-DiD design finalisation; matching; DiD estimation; robustness checks; methods note.	~20 days, concentrated in M3-M4, M7-M8, M34-M36	Short-term consultancy
Qualitative researcher	Scoping IDIs; endline IDIs, FGDs, KIIs; thematic and MSC analysis; qualitative report.	~20 days, concentrated in M1-M2 and M31-M37	Short-term consultancy
Female facilitator	Women-only focus groups; women's IDIs; analysis of gendered access data.	~8 days during M31-M34	Sub-contracted to qualitative researcher
Field enumeration team	Household survey administration in Pokot/Tugen at baseline and endline waves.	4-6 enumerators × ~18 days each per wave	Locally recruited via county M&E network
Field supervisor	Day-to-day enumeration management; back-checks; quality audits during fieldwork.	~40 days across both waves	Locally recruited

Two features of the team composition warrant brief explanation. The first is the deliberate separation of the external specialist role from the qualitative researcher role. These are different methodological domains with different analytical traditions, and the candidate pool for each is largely distinct. Combining the two into a single 'lead evaluator' role would either compromise the quality of one strand or substantially increase the day rate required to secure someone credible in both. The split arrangement is more cost-efficient and methodologically stronger.

The second is the inclusion of the female facilitator as a separately identified role rather than as an undifferentiated part of the qualitative team. This reflects the substantive importance of the women-only focus group discussions specified in Section 5 and the practical reality that securing a female facilitator with appropriate language skills and cultural competence for ASAL settings is a specific recruitment task that should be planned and budgeted explicitly. Treating this as a default sub-contract of the qualitative re

searcher risks under-investment in the role at the moment of recruitment.

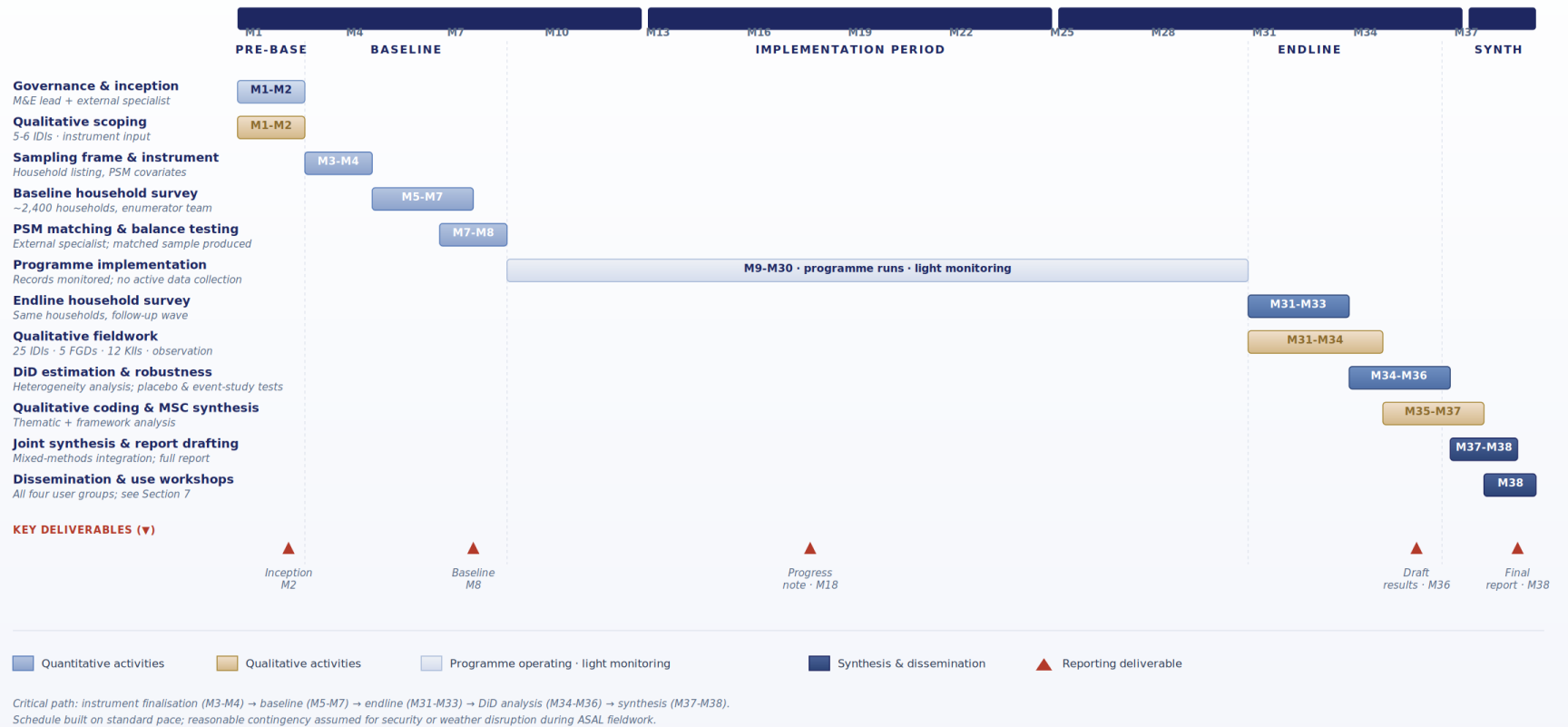
6.4 Activity schedule

The evaluation runs over 38 months from inception through dissemination. Figure 10 sets out the full activity schedule as a Gantt-style timeline showing the twelve major activity streams, the implementation period during which the programme operates without active data collection, and the five key reporting deliverables that mark milestones in the work.

FIGURE 10 • IMPLEMENTATION TIMELINE

Evaluation Activity Schedule and Key Deliverables

38-month evaluation timeline spanning pre-baseline scoping through final synthesis and dissemination.



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Figure 10. Implementation timeline showing activities, dependencies, and key deliverables across the 38-month evaluation period.

Source: Author's design, building on the sequencing framework introduced in Section 5.5.

The schedule's critical path runs through four sequential dependencies. Instrument finalisation in months 3-4 governs the start of baseline fieldwork; baseline fieldwork in months 5-7 must complete before PSM matching can produce the analytical sample; the implementation period must run its full course before endline fieldwork can begin in month 31; and endline data must be cleaned and matched before DiD estimation can proceed. Each dependency carries scheduling risk that the implementation arrangements must accommodate. The 38-month total includes a four-week contingency window built into the endline fieldwork phase to absorb the most likely sources of delay — security incidents in ASAL areas, seasonal disruption to fieldwork, or pastoralist mobility patterns that displace households during the planned data collection window.

Two design choices in the schedule warrant note. The first is the concentration of the main qualitative fieldwork at endline rather than throughout the implementation period, a choice developed in Section 5.5. The schedule reflects this: months 31-34 carry concentrated qualitative activity, while the long implementation phase from month 9 to month 30 carries no active data collection. The second is the placement of the joint synthesis and report drafting in the final two months (months 37-38). This compressed timeline is intentional: synthesis benefits from being conducted with both quantitative and qualitative results fresh, and an extended drafting period tends to produce a longer rather than a better report. The compressed window forces analytical discipline.

6.5 Budget

The indicative budget for the full evaluation is approximately USD 58,700. Table 9 sets out the cost categories, the calculation basis for each, and the indicative figure. The figures should be read as indicative only — actual costs will depend on consultant availability and day rates at the time of contracting, fieldwork conditions during the relevant period, ethics-approval timelines, and the exchange-rate environment between USD and KES. A budget refresh at the inception stage is essential.

Table 9. Indicative budget by cost category

Cost category	Calculation basis	Indicative cost (USD)
External specialist consultancy	20 days × \$550/day	11,000
Qualitative researcher consultancy	20 days × \$400/day	8,000
Female facilitator	8 days × \$300/day	2,400
Field enumerators (both waves)	5 enumerators × 18 days × 2 waves × \$25/day	4,500

Field supervisor	40 days × \$80/day	3,200
Field logistics & transport	Vehicle hire, fuel, accommodation across 22-24 villages × 2 waves	12,000
Translation & interpretation	Pokot/Tugen translators for survey and qualitative work	3,500
Data collection software (CAPI)	SurveyCTO or similar; licensing across 2 waves	1,800
Ethics approvals & registration	KEMRI/SERU fees; NACOSTI permits; ethics committee consultations	2,500
Dissemination & workshops	User-group workshops, county presentation, summary brief production	4,500
Contingency (10%)	ASAL fieldwork, security or seasonal disruption	5,300
TOTAL INDICATIVE		58,700

Three observations about the budget structure are worth flagging for the user of this design. The first is that consultancy costs (external specialist plus qualitative researcher plus female facilitator) account for approximately 36 per cent of the total. This is the analytical core of the work and the area where compromise on day rate or duration produces compromise on the credibility of the resulting findings. The implementing agency's CSR director should approach this category as a fixed analytical investment rather than a discretionary line to be reduced under budget pressure.

The second is that field logistics — vehicle hire, fuel, accommodation, per diems for the enumeration team across twenty-two to twenty-four villages over two waves — account for approximately 20 per cent of the total. This is the structural cost of conducting an evaluation in ASAL geography, and it does not compress under budget pressure without compromising the quality of fieldwork. A reduced logistics budget produces reduced contact between enumerators and respondents, which produces higher missing-data rates, which produces lower analytical credibility.

The third is the explicit 10 per cent contingency line. ASAL fieldwork is subject to a range of disruptions that cannot be fully predicted at the design stage — security incidents, seasonal disruption, equipment failures, ethics-approval delays. A budget without a contingency is a budget that will need to be renegotiated mid-evaluation; the explicit allocation signals professional realism about the operational environment.

Compared to a fully externally-led equivalent evaluation — which would typically run to USD 100,000-150,000 for an intervention of this scale — the hybrid model represents substantial cost efficiency. The

trade-off is in the M&E lead's time commitment, which is absorbed into the implementing agency's existing M&E function rather than added as a marginal evaluation cost.

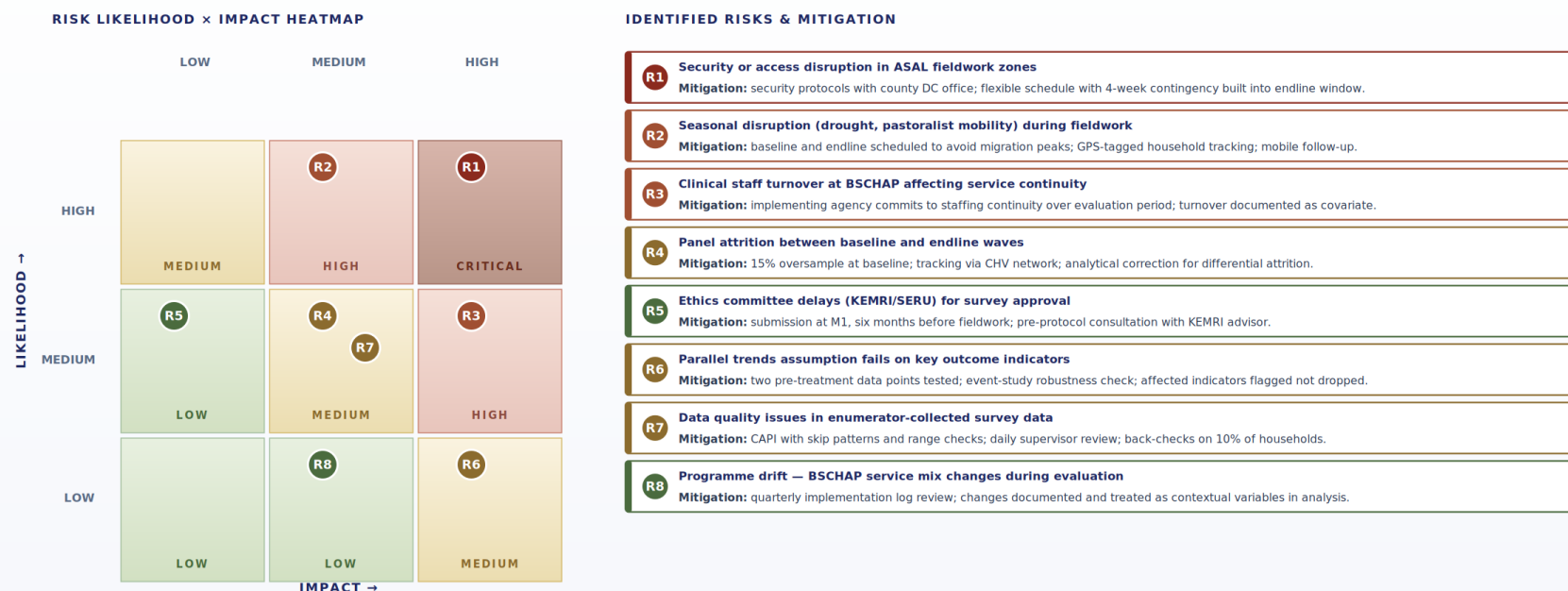
6.6 Risk register

Eight risks have been identified through structured review of the design, the implementation environment, and lessons from comparable evaluations in similar settings. Figure 11 presents the risk register, plotting each risk on a likelihood-impact heatmap alongside its mitigation strategy.

FIGURE 11 · RISK REGISTER

Risk Assessment with Mitigation Strategy

Eight identified risks plotted by likelihood and impact; each paired with a specific mitigation responsibility.



KEY MESSAGE

Three risks (R1, R2, R3) sit in the high-impact zone and warrant active monitoring. Five risks are low-to-medium and managed through standard evaluation practice.

Risk reviews are conducted at Months 2, 8, 18, and 30 by the M&E lead and external specialist jointly, with formal escalation to the implementing agency's CSR director where mitigation triggers are met.

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Figure 11. Risk register — eight identified risks with likelihood-impact assessment and mitigation strategies.

Source: Author's design, drawing on standard risk management practice in development evaluation.

Three risks sit in the high-impact zone and warrant active monitoring throughout the evaluation. Security or access disruption in ASAL fieldwork zones (R1) is the most consequential. The Pokot and Tugen catchment communities sit in an area that has historically experienced periodic inter-communal tension and where road access deteriorates during the long rains. The mitigation is multi-layered: security protocols agreed with the county Deputy Commissioner's office, flexibility in the fieldwork window to allow rescheduling around security incidents, and a four-week contingency built into the endline schedule. Where security materially worsens, the evaluation can be paused without compromising the analytical design, because the panel structure of the household survey means a delayed endline still produces valid DiD estimates.

Seasonal disruption (R2) is closely related but operationally distinct. Pastoralist mobility patterns mean that some households move with livestock to seasonal grazing areas during specific months of the year. Baseline and endline are scheduled to avoid migration peaks (typically the long dry season from January to March and the short dry season in September), GPS-tagged household locations enable tracking when households relocate, and mobile follow-up via SMS-based survey supplement is used where in-person follow-up is impossible.

Clinical staff turnover at BSCHAP (R3) affects the evaluation indirectly but materially. If the clinical officer or nurse leading the clinic departs during the evaluation period and is not replaced quickly, the intervention effectively changes — the service mix evaluated is not the service mix delivered. The implementing agency has committed in principle to continuity of clinical staffing over the evaluation period, but discretion remains with the agency's HR function. The mitigation is documentary: any turnover is logged with dates and continuity arrangements, and the turnover variable is included as a contextual covariate in the heterogeneity analysis. Where turnover proves substantial, the evaluation interprets findings with explicit reference to the staffing pattern that actually obtained.

Five further risks are managed through standard evaluation practice. Panel attrition (R4) is addressed through a 15 per cent oversample at baseline and analytical correction for differential attrition at endline. Ethics committee delays (R5) are addressed through early submission and pre-protocol consultation. The parallel trends assumption failing on key indicators (R6) is addressed through two pre-treatment data points and an event-study robustness check; affected indicators are flagged for cautious interpretation rather than dropped. Data quality issues (R7) are addressed through computer-assisted personal interviewing with skip patterns and range checks, daily supervisor review, and back-checks on a 10 per cent sample. Programme drift (R8) is addressed through quarterly implementation log review and treatment of documented changes as contextual covariates in the analysis.

Risk reviews are conducted at four formal points during the evaluation: at month 2 (post-inception), month 8 (post-baseline), month 18 (mid-implementation), and month 30 (pre-endline). At each review,

the M&E lead and external specialist jointly reassess the likelihood and impact of each identified risk, identify any new risks, and document the status of mitigation activity. Where any risk's likelihood-impact assessment moves into the critical or high-impact zone, formal escalation to the CSR director through the steering committee is triggered.

6.7 Quality assurance

Quality assurance operates at three layers. The first is data collection quality, where the focus is on minimising measurement error and maximising the reliability of the data on which the analytical work depends. The household survey instrument is administered using computer-assisted personal interviewing software with built-in skip patterns, range checks, and consistency validation. A field supervisor accompanies the enumeration team for the first two days of each wave, conducts back-checks on a 10 per cent random sample of completed interviews, and reviews daily upload logs to identify any enumerator producing outlier patterns. Where back-check discrepancy rates exceed 5 per cent, the relevant interviews are flagged for re-administration.

The second is analytical quality. The external specialist's analytical work is documented in a methodological log that records each consequential analytical choice — covariates included in the propensity score model, calliper width for matching, specification choices in the DiD regression, robustness checks performed — and the reasoning behind each. The log forms an annex to the final report and is available to any subsequent reader who wishes to interrogate the analytical decisions. This is the equivalent of a pre-analysis plan combined with a record of deviations from it, the documentation discipline that distinguishes credible impact evaluation from analytical opacity.

The third is reporting quality. Each major deliverable — inception report, baseline report, draft results, final evaluation report — is reviewed by the M&E lead and at least one independent reviewer (typically the independent steering committee member) before submission to the implementing agency. Findings that bear on the credibility of the broader geothermal project's social licence — particularly any negative findings on community-implementing-agency relations from the qualitative strand — receive specific independent review to ensure that the evaluator's analytical independence is visible in the reported text. The implementing agency reserves the right to comment on factual accuracy but not on analytical findings.

The implementation framework set out in this section is intended to be operationally realistic: the team is recruitable, the budget is defensible, the timeline accommodates the most likely disruptions, and the quality assurance arrangements are proportionate to the analytical credibility the evaluation needs to deliver. The framework should be refreshed at

the inception stage when actual day rates, ethics-approval timelines, and fieldwork conditions are known.

Section 7: Dissemination and Use

7.1 From evidence to use

An impact evaluation that produces credible findings but fails to reach the people whose decisions those findings should inform has not failed analytically — it has failed at the one thing that justifies the cost of conducting it. This is a more common failure mode in development evaluation than weak methodology. Evaluations are commissioned, conducted to a high technical standard, and then filed: a PDF on an organisational intranet, a slide deck presented once at a steering committee meeting and never referenced again, a finding that never reaches the operational manager who could act on it.

The dissemination and use strategy set out in this section is designed against that failure mode. It rests on a simple premise: different user groups need different evidence in different forms at different times, and a single generic final report cannot serve all of them well. The CSR department needs operational detail to refine the programme; senior management needs a headline finding to inform portfolio decisions; the county health department needs policy-relevant findings framed in terms of public health planning; the international development M&E community needs methodological transparency to apply the lessons elsewhere. Treating dissemination as the production of tailored outputs for defined audiences, rather than the publication of one document, is the central design choice in this section.

7.2 The reporting cycle

Table 10 sets out the five primary reports produced over the evaluation's 38-month cycle, their timing, length, and purpose. Figure 12 on the following page presents the same information as a visual timeline, integrated with the four user-targeted outputs that follow from the final report.

Table 10. Primary evaluation reports across the implementation cycle

Report	Timing	Length	Primary purpose
Inception report	Month 2	~15-25 pages	Confirms refined design after scoping; secures steering committee sign-off before fieldwork begins.
Baseline report	Month 8	~30-40 pages	Documents baseline sample characteristics, balance between treated and matched comparison groups, and any deviations from the design.

Progress note	Month 18	~5-8 pages	Brief internal update on implementation status, attrition, and risk register during the long implementation period.
Draft results	Month 36	~50-70 pages	Full quantitative and qualitative findings for independent review before the final report is finalised.
Final evaluation report	Month 38	~60-80 pages	Definitive synthesis with policy recommendations and the methodological log; the public record of the evaluation.

Each report has a distinct audience and purpose, and the design deliberately avoids treating later reports as expansions of earlier ones. The inception report is a design-confirmation document for the steering committee, not a preview of findings. The baseline report is a methodological transparency record — it documents the sample that was actually achieved and the balance between treatment and comparison groups, which matters for any reader assessing the credibility of the eventual causal claim. The progress note is intentionally brief: its purpose is to keep internal stakeholders informed during the long implementation period when no active data collection occurs, not to anticipate findings. The draft results report is the full analytical product, circulated for independent review before anything is finalised. The final evaluation report is the public record of the evaluation, and is the document from which every user-targeted output is derived.

FIGURE 12 · REPORTING CYCLE

Evaluation Deliverables Across the 38-Month Cycle

Five primary reports and four user-targeted briefs structure the flow of evidence from inception through use.



DERIVED FROM THE FINAL REPORT — FOUR USER-TARGETED OUTPUTS (M38)



DESIGN PRINCIPLE Each report is written **for a defined reader and a defined decision**, not as a generic deliverable. Length, structure, and tone are calibrated to use, not to convention.

The full final report is the definitive analytical record; the four briefs translate it into action-ready evidence for the user groups identified in Section 1.6.

Reports flow forward in time but findings flow backward into design: lessons from each report inform refinements to subsequent fieldwork and reporting.

Figure 12. Reporting cycle — five primary reports across the 38-month evaluation, with four user-targeted outputs derived from the final report.

Source: Author's design, drawing on report-format conventions introduced in coursework and standard 3ie/DAC evaluation reporting practice.

7.3 Dissemination strategy by user group

Section 1.6 identified four intended users of the BSCHAP impact evaluation: the implementing agency's CSR Department, the senior management of the parent organisation, the Baringo County Government Department of Health, and the international development M&E community working in arid and semi-arid lands. Figure 13 maps how findings reach each of these groups — the specific decisions each output is designed to inform, and the channel through which it is delivered.

The dissemination strategy follows a simple structural logic. A single evidence base — the final report, its methodological log, and its data appendices — sits at the centre. Each user group receives a distinct output derived from that evidence base, calibrated in length, tone, and content to the decisions the group actually faces. This is the same logic that governs effective communication in any technical field: the underlying analysis is singular and rigorous; its presentation is plural and audience-specific.

FIGURE 13 • DISSEMINATION MAP

How Findings Reach the Four User Groups

Each user group receives findings through tailored outputs and channels matched to their decision context.



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Figure 13. Dissemination map — how the evidence base reaches each of the four user groups through tailored outputs and channels.

Source: Author's design, building on the user-group analysis introduced in Section 1.6.

CSR Department

The CSR department is the most operationally proximate user group and the one with the most granular information needs. The department receives the full final report and a six-page operational brief presenting effect estimates with confidence intervals, service-mix recommendations, cost-effectiveness comparisons against relevant benchmarks, and heterogeneity findings by sub-group and distance band. Delivery includes an in-person half-day workshop at the implementing agency's M&E unit, structured around a working session on the recommendations rather than a one-way presentation. A quarterly follow-up over the year after delivery tracks whether and how recommendations are being implemented — a discipline that closes the loop between evaluation and action, and one that most evaluations omit.

Senior Management

Senior management — the parent organisation's board and chief executive — requires the shortest, most decision-oriented output: a two-page executive brief leading with the headline magnitude finding, framed explicitly in terms of social-licence implications and cost-per-life-touched comparisons that this audience uses to benchmark CSR investment against alternative uses of the same budget. A fifteen-minute board presentation with structured time for questions accompanies the brief. Selected findings are incorporated into the parent organisation's annual CSR report, which is frequently the only channel through which board-level stakeholders engage with CSR programme detail at all. The brief is written assuming the reader has not read the full report and will not read it subsequently — every essential fact must be present in the two pages.

County Health Department

The Baringo County Government Department of Health engages with the evaluation from a public health planning perspective rather than a programme management one. The relevant output is a four-page policy brief, co-signed with the County Director of Health to signal joint ownership of the findings rather than an external assessment of county-adjacent activity. The brief addresses county health-facility planning implications for ASAL areas, the practicalities of integrating BSCHAP into the county referral network, and lessons that would inform county-led replication of the model in other underserved sub-counties. A presentation at the county health forum, alongside other county-level health programme updates, situates BSCHAP within the county's broader health system rather than presenting it as an isolated CSR initiative. A joint M&E follow-up arrangement supports any county-led replication that follows.

INGO Practitioners

The international development M&E community engaged in arid and semi-arid lands programming is the user group most interested in methodology rather than findings about BSCHAP specifically. The output for this group is a practitioner note of approximately six pages, focused on the design lessons of broader applicability: the household-level reframing that rescues statistical power in small-catchment evaluations, the integration of PSM-DiD with a convergent parallel qualitative strand, and the practical realities of conducting rigorous evaluation in ASAL fieldwork conditions. This note is published openly on a repository such as Zenodo, accompanied by a webinar delivered through an established M&E community of practice, and supported by code and data

appendices that allow other practitioners to adapt the approach to their own contexts. This is also the output most directly relevant to the evaluator's own professional development: a published, citable practitioner note is a portfolio asset in its own right, distinct from the design document that produced it.

7.4 Use plan

Dissemination becomes use only when evaluation findings connect to specific forthcoming decisions with identifiable timing. Table 11 sets out, for each user group, the precise decision the evaluation is intended to inform and when that decision typically arises in the user's own planning cycle.

Table 11. Use plan by user group

User group	Specific decision the evaluation informs	Timing relative to decision
CSR Department	Whether to adjust the service mix (e.g. invest further in CHV outreach if uptake barriers are confirmed) and whether to extend or modify catchment boundaries.	Annual CSR planning cycle, typically Q1 of the following financial year.
Senior Management	Whether to replicate the BSCHAP model at other geothermal project sites within the parent organisation's portfolio.	Triggered by the next greenfield project's CSR planning phase.
County Health Department	Whether and how to integrate BSCHAP into the county's primary healthcare referral network and county health planning.	County Integrated Development Plan (CIDP) review cycle.
INGO Practitioners	Methodological choices for evaluation design in comparable ASAL, CSR-funded, or small-catchment health interventions.	Ongoing — practitioner note remains a standing reference.

The timing column matters as much as the content column. A finding delivered after the decision it was meant to inform has already been made has no use value regardless of its analytical quality. The evaluation's 38-month timeline was set, in part, with this constraint in view: the final report and its derived outputs are timed to land before the implementing agency's annual CSR planning cycle, giving the CSR department a realistic window to incorporate findings into the following year's planning rather than the year after.

7.5 Closing reflection on the evaluation design

The seven sections of this design document trace a single line of reasoning from intervention selection through to use. Section 1 established the programme and reframed the evaluation question from a binary test of whether BSCHAP works to a magnitude-estimation question about how much it works, for whom, and at what cost — a reframing made necessary by the ethical reality that withholding a known-beneficial health service for the sake of methodological purity is not a legitimate option. Section 2 developed the theory of change and exposed the assumption structure on which the causal claim depends, with particular attention to the behavioural and uptake

assumptions that distinguish a supply-side health intervention from a guaranteed behavioural outcome. Section 3 assessed three experimental identification strategies and found each unworkable for reasons that were substantive rather than merely inconvenient — ethical impermissibility, statistical underpowering at low numerosity, and conceptual misunderstanding of what randomised promotion actually randomises.

Section 4 developed the quasi-experimental alternative, and the household-level reframing introduced there is the single most consequential methodological decision in the entire design: it is what makes the difference between an evaluation that can detect a plausible effect and one that cannot. Section 5 integrated a qualitative strand that does not merely supplement the quantitative design but performs analytical work the quantitative design structurally cannot — explaining mechanism, testing assumptions, triangulating findings, and capturing consequences outside the indicator set. Sections 6 and 7 then established that the design is not only methodologically sound but operationally deliverable and use-oriented: a team that can be recruited, a budget that is defensible, a timeline that accommodates real-world disruption, and a dissemination strategy that connects findings to the decisions they are meant to inform.

Two limitations of the design deserve acknowledgement at this closing point, not because they were missed earlier but because a design document's credibility depends on naming its own boundaries explicitly rather than leaving them to be discovered by a critical reader. The first is that the entire design remains, as of this document, untested against real data. Every power calculation, every covariate balance expectation, every minimum detectable effect in Table 5 is a planning estimate. The actual analytical work — and the credibility that comes only from having executed a design rather than merely proposed one — remains to be done. The second is that no quasi-experimental design, however carefully constructed, achieves the identification certainty of a well-powered randomised controlled trial. The PSM-DiD strategy developed here is the strongest design available given the ethical and statistical constraints of the BSCHAP context, but it is not, and does not claim to be, equivalent in identification strength to the experimental designs that Section 3 ruled out.

This document is an impact evaluation design, not an impact evaluation. It demonstrates how a credible counterfactual can be constructed for a small-catchment, CSR-funded health intervention where randomisation is neither ethical nor statistically viable, and it sets out the operational and dissemination architecture needed to deliver that design as a piece of professional evaluation practice. The next step in this body of work — applying the design to data, whether real or credibly simulated — is what would transform this design into an evaluation with findings to report.

TERMS OF REFERENCE

Impact Evaluation of the Baringo-Silali Community Health Access Programme (BSCHAP)

Issued by: CSR Department, the implementing agency

Location: Baringo County, Kenya

Document status: Final draft, May 2026

Evaluator's note *This Terms of Reference is presented in the institutional voice of a commissioning body, as a ToR would be issued in practice. It was developed by Eric Kizambo Kachila as a portfolio artefact accompanying the BSCHAP Impact Evaluation Design Document, and demonstrates the capacity to produce the commissioning-side document that initiates an evaluation, as distinct from the evaluator-side design document that responds to it. The implementing agency is anonymised consistent with the companion design document; the programmatic and geographic context is retained for analytical realism. All figures are illustrative.*

1. Background and Rationale

1.1 The programme

The Baringo-Silali Community Health Access Programme (BSCHAP) is a Corporate Social Responsibility initiative of the implementing agency, operated at its geothermal project site in Baringo County, Kenya. The implementing agency established a staff medical clinic to provide occupational health services to its workforce and, recognising the acute health-access gap faced by neighbouring communities, extended the clinic's services to surrounding pastoralist communities free of charge. BSCHAP delivers free primary healthcare outpatient consultations, maternal and child health services, immunisation, basic diagnostics, and referral to communities that previously had no access to any formal health facility.

The programme's objectives are to improve access to primary healthcare, reduce preventable morbidity, improve maternal and child health outcomes, and reduce the financial burden households face when seeking care elsewhere. The intended ultimate outcome is improved community health and wellbeing among the surrounding pastoralist population.

1.2 Rationale for the evaluation

Despite operating for several years, BSCHAP has never been formally evaluated. The implementing agency's CSR Department requires credible evidence of the programme's impact to inform resourcing decisions, to demonstrate accountability to senior management and the Board, and to contribute to the broader evidence base on CSR-funded health models in remote settings. Because access to primary healthcare is already well established in the public health literature as beneficial, this evaluation does not ask whether the programme works in principle. It seeks instead to estimate the magnitude of BSCHAP's causal impact and to understand the mechanisms through which that impact, if present, operates.

The overarching evaluation question that this assignment is commissioned to answer is: to what extent, and through what mechanisms, has BSCHAP causally improved health-seeking behaviour and health outcomes among communities surrounding the project site, compared with similar communities without access to the programme?

1.3 Context

The Baringo-Silali area is among Kenya's most remote and underserved semi-arid zones: sparsely populated, with poor road infrastructure and minimal government services. The nearest public health facility outside the BSCHAP catchment is several hours away by road. Communities in the catchment are predominantly Pokot and Tugen pastoralists, with low household incomes and historically limited contact with formal health systems. An estimated 500 to 800 unique individuals access the BSCHAP clinic annually, drawn from approximately 22 to 25 villages within roughly 30 kilometres of the project site.

1.4 Stakeholders

The implementing agency's CSR Department funds and implements the programme; day-to-day clinic operations are supervised by the site medical officer. Community Health Volunteers and village elders support community mobilisation and patient referral. The County Government of Baringo's Department of Health receives patients referred from BSCHAP for services beyond the clinic's scope. The evaluation has four intended user groups: (1) the implementing

agency's CSR Department, (2) the implementing agency's senior management, (3) the County Government of Baringo, and (4) external development partners with an interest in CSR-funded health service models in arid and semi-arid contexts.

1.5 Prior evaluation and available framework

No prior formal evaluation of BSCHAP exists. A theory of change, results framework, evaluation questions, and indicator framework have been developed in preparation for this assignment and are attached as Annex A. The selected evaluation team is expected to review, validate, and where necessary refine this framework during the inception phase.

2. Purpose of the Evaluation and Evaluation Questions

2.1 Purpose and intended use

This evaluation serves three purposes. The first is accountability: providing the implementing agency's senior management and Board with credible evidence on whether the CSR investment in BSCHAP delivers measurable benefit to surrounding communities. The second is programme improvement: informing operational decisions on clinic resourcing, service mix, and outreach intensity. The third is organisational and sector learning: generating evidence on CSR-funded primary healthcare models in remote, underserved settings that is transferable to other contexts, including other project sites operated by the implementing agency and to the wider community of development practitioners working in arid and semi-arid lands.

2.2 Evaluation criteria

Impact is the lead evaluation criterion. Two complementary criteria are also addressed: cost-effectiveness (the magnitude of impact achieved per unit of CSR expenditure, an important consideration given the programme's small catchment population) and relevance (the extent to which the services delivered match the health needs and priorities of the surrounding communities). Effectiveness and sustainability are noted as secondary considerations but are not the primary focus of this assignment.

Each evaluation question in Table 1 below is mapped to the criterion it primarily serves. The impact criterion is addressed at all three levels of the results chain — immediate, intermediate, and ultimate — since the evaluation's lead concern is the magnitude of BSCHAP's causal effect at each stage of the pathway from access to outcome. The cost-effectiveness criterion is addressed specifically through the intermediate and ultimate questions on healthcare transport expenditure and catastrophic health expenditure, since these indicators speak directly to the financial burden the programme is intended to relieve relative to its cost. The relevance criterion is not assessed through a discrete quantitative indicator in Table 1; it is addressed through the qualitative strand specified in Section 3.2, which examines whether the services BSCHAP delivers correspond to the health priorities communities themselves identify.

2.3 Evaluation questions and indicators

Table 1 sets out the evaluation questions at three levels of the results chain, together with the indicators against which each will be assessed. The full theory of change and indicator rationale are provided in Annex A.

Table 1. Evaluation questions and indicators

Level	Evaluation question	Indicators	Criterion addressed
Immediate	Does BSCHAP increase formal care-seeking, immunisation coverage, and antenatal care completion?	% households seeking formal care when last ill; % children 12-23 months fully immunised; % pregnant women completing 4+ ANC visits	Impact
Intermediate	Does BSCHAP reduce illness burden, healthcare transport costs, and obstetric complications?	Average illness days per household/month; average monthly healthcare transport expenditure; % deliveries without documented obstetric complication	Impact; Cost-effectiveness
Ultimate	Does BSCHAP reduce catastrophic health expenditure and increase skilled birth attendance?	% households with catastrophic health expenditure (>10% of income); % births attended by skilled personnel	Impact; Cost-effectiveness

Under-5 mortality is noted for completeness but is not treated as a primary causal outcome indicator: the catchment population is too small to support a statistically powered causal test on a rare-event outcome of this kind. The selected evaluation team should report on under-5 mortality descriptively and contextually only.

3. Approach and Methodology

3.1 Overarching methodological framework

The evaluation shall use a mixed-methods quasi-experimental design, combining a quantitative difference-in-differences (DiD) approach strengthened by propensity score matching (PSM) at household level with a complementary qualitative strand. Bidders proposing an alternative methodology must provide an explicit justification, referencing the constraints set out below, and demonstrate that the alternative offers superior identification under those same constraints.

Three experimental designs were assessed during the evaluation's design phase and are documented in Annex B: randomised assignment, randomised phase-in, and randomised promotion. Randomised assignment and phase-in were found unsuitable on both ethical and statistical grounds: they would require withholding or delaying access to a known-beneficial health service from communities with no alternative facility, and the programme's small catchment (approximately 22 to 25 villages) is too small a number of units for randomisation to reliably balance treatment and comparison groups. Randomised promotion was found conditionally viable and is retained as a supplementary identification strategy, since it does not withhold the programme from any community — it varies only the intensity of demand-side outreach — and is specified further in Section 3.2 below. Regression discontinuity was assessed and rejected because no defensible discontinuous eligibility rule exists in the BSCHAP context.

The selected quasi-experimental approach is ethically appropriate because it does not require withholding or manipulating programme access: the comparison group is constructed from communities that do not access the clinic because of geographic distance, not because of deliberate exclusion by the evaluation. Propensity score

matching at household level, rather than village level, is required: matching the approximately 12 treated villages against approximately 10 to 12 comparison villages would produce an analytical sample of roughly 24 units at village level — too few to support credible inference about anything but very large effect sizes. The selected team must conduct matching and outcome analysis at household level, with appropriate cluster-robust standard errors at village level to account for the resulting non-independence of observations within villages.

3.2 Data collection and analysis

The quantitative strand requires household surveys administered at baseline and endline in both treatment villages (approximately 12 villages within 30 kilometres of the project site) and matched comparison villages (approximately 10 to 12 villages, 50 to 80 kilometres away, sharing the same livelihood system, ethnicity, and agro-ecological zone). The survey instrument must capture the indicators in Table 1, alongside the household-level covariates required for matching. Clinic registers held by the implementing agency are available to the selected team as a source of output-level data and as a triangulation reference for survey-reported attendance.

Where the selected team identifies scope to vary the intensity of demand-side outreach activity (such as Community Health Volunteer deployment or transport subsidy schemes) across villages during the assignment, the resulting variation in uptake should be exploited as a supplementary identification strategy using instrumental-variables estimation, with the Wald estimator — the ratio of the difference in outcomes to the difference in participation rates between higher- and lower-outreach villages — as the preferred form for a single-instrument specification. This supplementary analysis is intended to triangulate against the primary PSM-DiD estimate: convergence between the two would substantially strengthen the credibility of the evaluation's causal claims, since the two methods rely on different identifying assumptions. The selected team's technical proposal should set out how this supplementary strategy would be operationalised, including the proposed outreach variation and its anticipated effect on take-up.

The qualitative strand requires in-depth interviews, focus group discussions (including women-only groups), key informant interviews, and structured observation, addressing the mechanisms behind any quantitative findings, testing the implementation and uptake assumptions in the theory of change, capturing beneficiary experience, and surfacing unintended consequences of the programme. The qualitative strand also addresses the relevance criterion specified in Section 2.2 directly, by examining whether the service mix BSCHAP currently delivers corresponds to the health priorities communities themselves identify, and by documenting any unmet needs that fall outside the programme's current scope. The qualitative strand is integrated with the quantitative strand through joint display analysis at the synthesis stage, not conducted as a separate or subordinate exercise.

Bidders should note that baseline data for comparison villages will, in most cases, need to be reconstructed through recall-based survey methods, since no comparable prior data collection has occurred in those villages. This is a recognised limitation of the available data environment and proposals should set out how recall bias will be addressed.

3.3 Known limitations and required mitigation measures

- Small catchment and statistical power: with 500-800 annual beneficiaries from approximately 12 villages, power at village level is severely limited. The selected team is required to conduct analysis at household level, raising the analytical sample to approximately 2,000-2,400 households.

- ▶ **Parallel trends assumption:** the DiD design requires that treatment and comparison communities would have followed similar outcome trends absent BSCHAP. The selected team must validate this assumption using available pre-programme trend data and report explicitly on any indicators for which the assumption appears not to hold.
- ▶ **Unobservable confounders:** matching addresses observable differences only. The selected team's qualitative strand is expected to surface and interpret plausible sources of residual confounding, such as community leadership quality or pre-existing relationships with the implementing agency.
- ▶ **Recall and reporting bias:** the selected team must use bounded recall periods, triangulate self-reported survey data against clinic registers, and ensure enumerators are trained and fluent in the relevant local languages.
- ▶ **Spillover and contamination:** comparison villages selected must lie beyond practical travel range of the clinic, and the selected team must measure and adjust for any cross-over access where it is found to occur.

4. Deliverables and Reporting Schedule

Table 2 sets out the deliverables required under this assignment and the schedule against which they fall due. The assignment is structured over 38 months from contract signature to delivery of the final report and dissemination outputs.

Table 2. Deliverables and reporting schedule

Deliverable	Due	Format
Inception report	Month 2	~15-25 pages, including finalised methodology, sampling frame, instruments, and a dissemination plan (see Section 5)
Baseline report	Month 8	~30-40 pages, including sample characteristics, covariate balance tables, and the propensity score matching log
Progress note	Month 18	~5-8 pages, internal circulation, covering implementation status, attrition, and risk register update
Draft evaluation report	Month 36	~50-70 pages, full quantitative and qualitative findings, submitted for independent review
Final evaluation report	Month 38	~60-80 pages, definitive synthesis, policy recommendations, and methodological annex
Dissemination outputs (4)	Month 38	CSR operational brief, senior management executive brief, county health policy brief, INGO practitioner note — see Section 5

All deliverables are subject to review and approval by the implementing agency's M&E lead, with the draft evaluation report additionally subject to independent review before finalisation. The selected team's contract is structured so that final payment is contingent on delivery and acceptance of the final evaluation report and the dissemination outputs in Table 2, not on the inception or baseline reports alone.

5. Dissemination Requirements

Dissemination is a contractual deliverable under this assignment, not a discretionary activity to be addressed only once findings are available. The selected team is required to submit, as part of the inception report, an initial dissemination plan identifying the intended outputs and channels for reaching each of the four user groups named in Section 1.4: the implementing agency's CSR Department, the implementing agency's senior management, the County Government of Baringo's Department of Health, and the external development partner community. The inception-stage dissemination plan need not specify final detail, but must establish the structure, responsibilities, and budget allocation for dissemination activity, so that dissemination is appropriately resourced from the outset of the assignment rather than improvised once the evaluation concludes.

The inception-stage dissemination plan is expected to be revised as the assignment progresses, in light of evolving understanding of the most relevant findings, the stakeholders' evolving information needs, and any operational learning from the evaluation's qualitative fieldwork. A revised dissemination plan is required as part of the draft evaluation report submission at Month 36, reflecting this iterative refinement.

At minimum, the selected team is required to produce: a concise operational brief for the CSR Department, calibrated to inform service-mix and resourcing decisions; an executive brief of no more than two pages for senior management, calibrated to inform CSR portfolio and replication decisions; a policy brief for the County Government of Baringo, co-developed with the County Department of Health; and a practitioner-facing methodological note suitable for publication on an open repository, addressed to the external development partner community. Each output is to be delivered alongside an appropriate engagement activity — a workshop, briefing, or presentation — rather than as a standalone document with no accompanying discussion. The dissemination budget line in Section 6 below is to be treated as a fixed allocation for this purpose and is not available for reallocation to other cost categories without the explicit written agreement of the implementing agency's CSR Department.

6. Team Composition, Budget, and Timeline

6.1 Required team composition and qualifications

Table 3 sets out the roles required for this assignment and the minimum qualifications expected of bidders for each externally-contracted role. The implementing agency's own M&E function provides internal coordination and is not a role for which bidders are expected to propose personnel.

Table 3. Required team composition and qualifications

Role	Minimum qualifications required of bidders
Team leader / external specialist	Postgraduate degree in economics, statistics, or a related quantitative field; minimum 5 years' experience designing and conducting quasi-experimental impact evaluations; demonstrated proficiency in Stata or R; prior experience in East African or comparable ASAL contexts preferred.
Qualitative researcher	Postgraduate degree in social sciences, public health, or related field; minimum 4 years' experience in qualitative evaluation methods; demonstrated experience

	designing and leading focus group discussions and key informant interviews in pastoralist or similarly marginalised community contexts.
Female field facilitator	Fluency in Pokot and/or Tugen; demonstrated experience facilitating women-only group discussions on sensitive health topics; prior fieldwork experience in Baringo County or comparable ASAL settings preferred.
M&E lead (provided by the implementing agency)	Not a bidder requirement — the implementing agency's existing M&E function will coordinate, liaise, and provide quality assurance oversight throughout the assignment.

6.2 Indicative budget envelope

Table 4 sets out the indicative budget envelope for this assignment. Bidders are required to submit a detailed costed proposal against these categories; figures below are indicative planning estimates only and do not constitute a guaranteed contract value.

Table 4. Indicative budget envelope (USD)

Cost category	Indicative envelope (USD)
Consultancy fees (team leader, qualitative researcher, facilitator)	21,400
Field enumeration and supervision	7,700
Field logistics and transport	12,000
Translation, interpretation, and data collection software	5,300
Ethics approvals and regulatory compliance	2,500
Dissemination and use (see Section 5)	4,500
Contingency (10%)	5,300
TOTAL INDICATIVE ENVELOPE	58,700

6.3 Timeline

The assignment runs over 38 months from contract signature. The critical path runs from instrument finalisation in months 3-4, through baseline fieldwork in months 5-7, an implementation period of approximately 22 months during which the programme operates and no active data collection occurs, endline fieldwork in months 31-33, analysis in months 34-36, and synthesis and dissemination in months 37-38. A contingency allowance of four weeks is built into the endline fieldwork window to accommodate security, seasonal, or other operational disruption typical of fieldwork in this geography.

7. Governance and Quality Assurance

The implementing agency's CSR Director holds ultimate responsibility as the assignment's sponsor and approves the inception report, the budget, and the final evaluation report. The implementing agency's M&E lead serves as the assignment manager, coordinating the selected team's work, managing field operations logistics, and conducting quality assurance over data collection and reporting. A steering committee, comprising the CSR Director, the M&E lead, a representative of the County Government of Baringo's Department of Health, and an independent evaluation specialist external to the implementing agency, oversees the assignment at the inception, mid-point, and synthesis stages.

The selected team's analytical independence is a contractual term of this assignment: the team's methodological decisions and findings, once reached through the agreed methodology, are reported without alteration by the implementing agency. The implementing agency reserves the right to comment on factual accuracy in any draft report but not to require changes to analytical findings or conclusions. This term protects the credibility of the evaluation as the basis for the accountability and learning purposes set out in Section 2.1.

8. Proposal Submission Requirements

Bidders are required to submit a technical proposal addressing the methodological approach in Section 3, a detailed work plan against the deliverables schedule in Section 4, CVs for all proposed team members against the qualifications in Table 3, and a costed financial proposal against the budget categories in Table 4. Technical proposals are expected not to exceed 20 pages excluding annexes and CVs. The implementing agency will evaluate proposals on technical quality (60%), team qualifications and relevant experience in comparable contexts (25%), and cost (15%).

Annexes

- ▶ Annex A — Theory of Change, Results Framework, and Indicator Rationale
- ▶ Annex B — Assessment of Experimental and Quasi-Experimental Identification Strategies
- ▶ Annex C — Draft Household Survey Instrument (to be finalised at inception)

Note: Annexes A and B correspond to Sections 2-4 of the companion BSCHAP Impact Evaluation Design Document, prepared as part of the same body of work.